

What Adapter Ensembling Actually Buys: A Pre-Registered Audit, and the Scalar That Replaces the Population

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Abstract

Parameter-efficient fine-tuning leaves practitioners with populations of adapters, and a large literature reports that combining them helps. We ask what combining them actually buys once the single-adapter baseline is made strong. We pre-register and release ADAPTERFRONTIER, a 1,028-cell corpus registered at git SHA 02a5a6bc *before any adapter was trained* (67 pools, 13 base models, 8 tasks)¹, adjudicating every (pool, method, baseline, metric) cell with paired bootstrap CIs and corpus-wide BH-FDR control. Our single-adapter baseline is deliberately strong: measured from the released manifests it received a median $1.45\times$ the pool’s training compute (and up to $2.96\times$) plus a hyperparameter search the ensemble arm never gets. Against that baseline, encoder ensembling **never significantly improves accuracy** — 0 of 92 clean cells, identically before and after FDR correction, with 33.7% significantly *worse*. It does improve ECE on 52 of the same 92 cells, and we initially read that as a dissociation. It is not one. The comparison is against a single adapter chosen by validation *accuracy* and never calibrated, which is the weak-baseline shortcut this paper is otherwise about; on the pools where we can run the missing control, **one temperature fitted on held-out data erases the advantage** (ensemble better in 4/6 pools $\rightarrow$ 2/6, mean ΔECE -0.032), the multiclass Brier never corroborated it (34/92), and the mechanism is confidence shrinkage — which a scalar reproduces at $1\times$ compute instead of $N\times$. The practical consequence is to calibrate one adapter rather than serve N . We then price three evaluation shortcuts: selecting on test is worth $+1.12\text{pp}$ on average to whichever arm may peek, comparing against a same-size rather than a stronger single raises

SUPPORTED from 9.9% to 27.2% of cells, and reporting task means lets one -0.012pp “no effect” average conceal 37 significant cells. We release the cells, per-example predictions, split manifests and adjudication code so new methods can be compared without retraining.

1 Introduction

Parameter-efficient fine-tuning (PEFT) with LoRA-style adapters (Hu et al., 2022) has made it routine to keep dozens of task-specific adapters around: serving stacks such as vLLM and S-LoRA (Kwon et al., 2023; Sheng et al., 2024) route per request to one of N adapters at negligible extra cost, and frameworks such as LoRAHub (Huang et al., 2024) target multi-adapter composition directly. The practitioner’s question is whether *combining* those adapters — averaging weights, voting on labels, or routing through them — beats spending the same budget on a single adapter. A large literature reports that it does. We ask a prior question: *how much of that reported benefit survives a strict evaluation protocol?*

Answering this requires comparisons that were fixed before the data was seen, because the choices involved — which single adapter counts as the baseline, how it was selected, at what budget, and on which benchmark — are exactly the choices that determine the answer. We therefore pre-registered ADAPTERFRONTIER at git SHA 02a5a6bc *before any adapter was trained*, and adjudicate each (pool, method, baseline, metric) cell independently with paired bootstrap CIs, sign-flip tests, and corpus-wide Benjamini–Hochberg FDR (Benjamini and Hochberg, 1995).

A strong baseline, and a dissociation. The comparison that decides this question is the single adapter you would train instead. We make that competitor deliberately strong: our pre-registered `n_rank` baseline spends the pool’s budget on

¹Cell count: $1,028 = 827$ v1.0 main-pool cells (55 pools $\times$ ensemble methods $\times$ $\{\text{best_of_n, n_rank}\} \times \{\text{accuracy, ECE}\}$) $+ 100$ v1.1 HellaSwag multichoice cells $+ 101$ baseline self-comparison cells. The 2 v1.1 GSM8K and 6 v1.2 supplementary pools (App. K) are reported separately in Table 9 and are not in the headline 1,028.

adapter rank, and measured from the released manifests it in fact received a median $1.45\times$ the pool’s training wall-clock (max $2.96\times$; 47 of 48 pool–baseline pairs favour the baseline) together with a rank $\times$ epoch $\times$ seed search selected on held-out data that the ensemble arm never receives. It also serves one adapter per example against the ensemble’s N . It is therefore stronger than a matched competitor, and that cuts against us: a null against a better-resourced single is biased towards the null. The null does not weaken where the advantage is smallest (§4), but we state it against a single given *at least* the pool’s budget. Against it, on the clean encoder slice: **no cell shows a significant accuracy gain** (0 of 92, identically pre- and post-FDR) while 33.7% are significantly reversed (mean $\Delta = -1.27\text{pp}$). Our pre-registration declared accuracy and calibration co-equal arms, and on *the same 92 cells* ensembling improves ECE in 52. We read that as a dissociation until we noticed the comparison was not fair the way the accuracy arm’s was: that single adapter is chosen by validation *accuracy* and never calibrated. One temperature fitted on held-out data takes the ensemble from better-calibrated in 4/6 pools to 2/6 (§4). The honest statement is narrower and more useful: against a single given the pool’s budget, populations buy neither accuracy nor reliability that a scalar cannot, at $1\times$ compute instead of $N\times$.

Three shortcuts move the verdict. That null result coexists with a literature reporting gains, so we measure what closes the gap (§5). Selecting the “best single” on the test split is worth $+1.12\text{pp}$ on average to whichever arm is allowed to peek, and inflates 82% of pools. Comparing against a same-size single rather than a compute-matched one moves the SUPPORTED rate from 9.9% to 27.2%. Reporting one mean per task hides the structure entirely: MNLI’s -0.012pp average reads as “no effect” while concealing 37 significant cells pointing in both directions. Each shortcut is individually defensible-looking; together they are enough to flip the sign of the conclusion.

Contributions. (i) **A pre-registered corpus and adjudication framework:** 67 adapter pools and 1,028 cell JSONs with per-cell CIs, p -values and FDR-corrected verdicts, plus the locked pre-registration and amendment ledger; new methods are compared by submitting per-example predictions, with no retraining. (ii) **A null under a strong baseline, and the control that explains the one**

apparent exception: against a single adapter that received more training compute and a hyperparameter search, PEFT ensembling never significantly improves encoder accuracy (0/92) and frequently harms it. It does improve ECE on 52 of the same cells, but that comparison omits post-hoc calibration; supplying it removes most of the effect, which we trace to confidence shrinkage and replace with a single scalar. (iii) **A protocol-sensitivity analysis** pricing three shortcuts that each move the verdict (Table 3, Fig. 1). (iv) **Negative results reported as such:** the diversity–quality frontier predicts accuracy deltas out of pool (corpus-wide LOPO-CV $R^2=0.597$ over 48 pools, 95% CI $[+0.30, +0.71]$; encoder-only $R^2=0.616$ over 23 pools) but — because those deltas are predominantly negative — it predicts how much a pool stands to *lose*; it does not extend to calibration, and three candidate mechanisms are tested and refuted (App. W).

Descriptive regularities (oracle, not policy). Cells do differ, and the differences are legible as five rules D1–D5 (App. P). They were distilled from the same corpus they describe and evaluated on it, so they are an *oracle upper bound*, not a deployable policy (L11) — a prospective, validation-only version degenerated to always choosing the single adapter.

2 Related work

PEFT and adapter ensembling. The adapter line of work begins with bottleneck adapters (Houlsby et al., 2019) and the AdapterHub framework (Pfeiffer et al., 2020); LoRA (Hu et al., 2022), DoRA (Liu et al., 2024), PiSSA (Meng et al., 2024), AdaLoRA (Zhang et al., 2023), and VeRA (Kopiczko et al., 2024) make many low-rank adapters per task practical. Combination strategies split into three families: *prediction-space* (soft/hard voting, logit averaging (Lakshminarayanan et al., 2017)); *weight-space* (model soups (Wortsman et al., 2022), TIES (Yadav et al., 2023), DARE (Yu et al., 2024)); and *routing / mixture-of-adapters* (AdapterFusion (Pfeiffer et al., 2021) composes via attention; LoRAHub (Huang et al., 2024) optimizes a continuous weight vector; AdaMix (Wang et al., 2022) mixes multiple LoRA experts inside training; MoLE (Wu et al., 2024) and MoLoRA (Caccia et al., 2023) learn routing over adapter experts; HydraLoRA (Tian et al., 2024) uses asymmetric multi-head LoRA; MultiLoRA (Wang et al., 2023) reduces interference among

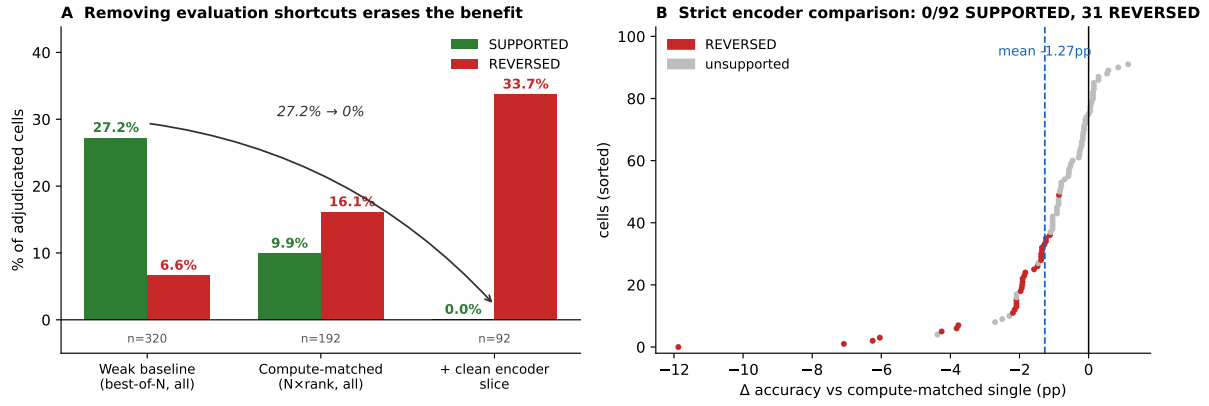


Figure 1: **The apparent ensembling benefit is a product of evaluation protocol.** (A) The fraction of cells adjudicated SUPPORTED collapses as shortcuts are removed: against the pool’s own best single (`best_of_n`) 27.2% of cells are SUPPORTED, against the pre-registered compute-matched single (`n_rank`) 9.9%, and on the clean encoder slice *none* are. REVERSED moves the other way (6.6% $\rightarrow$ 16.1% $\rightarrow$ 33.7%). (B) All 92 strict encoder cells, sorted, colored by post-FDR verdict: the mass lies left of zero (mean $-1.27pp$) and no cell is SUPPORTED.

co-trained LoRAs). Our adjudication framework treats each of these methods as a *plug-in* that produces per-example test predictions and can be added to our cell corpus without retraining the pools. We implement and report prediction-space methods directly (`soft_vote`, `logit_avg`, `majority_vote`, `greedy_soup`, plus a prediction-space `uniform_soup` proxy that averages per-adapter logits); weight-space TIES/DARE/uniform-merge implementations live in `weight_space_merge.py` and are reported as a separate family in App. C. The cell JSONs include the input schema needed to evaluate routing/MoE methods (per-pool logits cache).

Pre-registration in ML. Pre-registration is standard in clinical trials and increasingly common in cognitive science but rare in ML (Munafò et al., 2017; Loken and Gelman, 2017); Hofman et al. (2021) argue for tighter coupling of explanation and prediction in computational social science, an argument that translates directly to empirical NLP — Bowman and Dahl (2021) make a closely related case that NLU benchmark results need pre-committed protocols if they are to drive scientific (rather than leaderboard) progress. Dehghani et al. (2022) further argue that compute-matched comparisons are essential to honest efficiency claims — a principle we apply across every cell. We adopt pre-registration here: the protocol, splits, baselines, and statistical tests were locked at git SHA 02a5a6bc before any pool was trained.

Calibration of fine-tuned LMs. ECE in its modern form was introduced by Naeini et al. (2015); the equal-mass 15-bin histogram became standard via Guo et al. (2017), with refinements such as soft calibration objectives (Karandikar et al., 2021). Nixon et al. (2019) survey ECE estimator bias under varying bin counts and small- N regimes; Minderer et al. (2021) re-examine calibration of modern (post-2020) architectures and find that distribution-shift calibration does not always follow in-distribution calibration. Deep ensembles (Lakshminarayanan et al., 2017) reliably lower ECE on classifiers, but Ashukha et al. (2020) argue that in-domain uncertainty comparisons, ensembles included, should be made after temperature scaling, since an uncalibrated baseline confounds them, and Ovadia et al. (2019) find that temperature scaling’s in-distribution gains do not survive dataset shift, where ensembles hold up best. Our recalibration control applies the first point to PEFT pools; the second bounds it (L13). We extend this measurement to PEFT pools and report Brier, NLL, MCE alongside ECE, with paired bootstrap CIs; per-token logprob ECE for generative tasks (App I) is a non-standard adaptation we explicitly document.

Efficient multi-adapter serving. S-LoRA (Sheng et al., 2024) and Punica (Chen et al., 2024) (the latter integrated in vLLM (Kwon et al., 2023)) enable thousands of LoRA adapters to share a single base-model forward; we measure the runtime cost of cell-by-cell ensembling under this serving regime.

When reported gains do not survive fair comparison. Several fields have found that a literature of reported improvements shrinks or disappears once baselines are tuned and budgets are matched: GAN variants become indistinguishable under equal hyperparameter search (Lucic et al., 2018), deep metric-learning gains are “marginal at best” under controlled evaluation (Musgrave et al., 2020), and most neural recommenders are beaten by simple heuristics (Ferrari Dacrema et al., 2019). The recurring cause is not fraud but protocol: an under-tuned or under-resourced baseline, a selection step that sees the test set, or an aggregate score that hides variance (Dodge et al., 2019; Reimers and Gurevych, 2017). NLP has documented the same failure modes for benchmarking practice (Bowman and Dahl, 2021) and for efficiency claims specifically, where comparisons that do not fix the compute budget can invert conclusions (Dehghani et al., 2022). Two literatures bear directly on our question. PEFT evaluation has already had its reality check: Chen et al. (2022) find validation and testing practice across PEFT studies unreliable once method instability is accounted for. And *ensemble versus one bigger model at matched cost* is an open dispute rather than a naive assumption: Abe et al. (2022) find a single larger network replicates ensemble gains with diversity contributing little, Kondratyuk et al. (2020) conclude the opposite at matched FLOPs, Lobacheva et al. (2020) show the optimum depends on budget, and Theisen et al. (2023) predict the gain from the disagreement-to-error ratio. Our accuracy result is the PEFT instance of the Abe et al. side; our frontier is consistent with Theisen et al.. We claim priority for neither. Ours is the pre-registration before training, the cell-level adjudication, and the per-shortcut price in percentage points (§5).

Benchmarks for NLP reasoning. HellaSwag (Zellers et al., 2019), GSM8K (Cobbe et al., 2021), and the GLUE-family tasks (Wang et al., 2018) are scored by our own evaluators, not by lm-evaluation-harness (Gao et al., 2023), though we follow its length-normalised log-likelihood convention on HellaSwag. We use them as cells, not as headline numbers.

3 Benchmark composition

3.1 Pool types

Per (model, task) cell:

- **Pool-A (seed-only):** 20 LoRA adapters with fixed ($r = 8, \alpha = 16, lr = 3e-4, target = qv$), seed varied across $\{11, 21, \dots, 201\}$.
- **Pool-B (HP-diverse):** 24 adapters from $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{1e-4, 3e-4\}$.
- **Pool-C (method-mixed):** 20 adapters, $\{\text{LoRA, DoRA, rsLoRA, PiSSA}\} \times 5$ seeds.

Counting pools. The release holds 99 manifests: 73 trained ensemble pools, 25 n_rank baseline pools and one smoke test. The pre-registered corpus is 67 of the 73 (55 v1.0 classification + 12 v1.1 HellaSwag/GSM8K); the other 6 are v1.2 supplements (Table 9). Where a table counts 80 pools, it is the 55 v1.0 pools plus the 25 baselines, which also enter best_of_n comparisons. Realized sizes fall short of the design (20/24/20): median 19, range 7–24; baselines run 6–20, so best-of- N selection pressure varies by cell.

Compute matching. Per-pool grids are in App. J. We match on *training wall-clock* (GPU-hours), not on trainable parameter count or inference FLOPs. The n_rank baseline ($r \in \{16, 32\} \times \text{epochs} \in \{2, 4\} \times 5$ seeds, best-of- N selected on val_selection) was *designed* to consume roughly the GPU-hours of a 20-adapter Pool-A sweep (20 runs at $r \in \{16, 32\}$ instead of $r=8$, same token budget per adapter); as realized it consumed a median $1.45 \times$ (§4). Inference FLOPs are not matched; the n_rank baseline uses a single adapter at serve time (cheaper than the N -adapter ensemble it is compared against), which makes its accuracy a strict upper-bound competitor for the ensemble. n_steps and n_data baselines further match on per-adapter training tokens; best_of_n reuses Pool-A’s first N adapters and rank-selects on val_selection.

Frontier definition (formal). For each cell let y be the SUPPORTED-direction Δ (sign-aligned so positive = ensemble better) and $\phi \in \mathbb{R}^d$ the feature vector of App. Z. The *diversity-quality frontier* is $\hat{y} = f(\phi)$ fit by leave-one-pool-out CV. We fit three classes per slice under identical folds (ridge, random forest, gradient boosting; settings and all three scores in Table 1) and report ridge — the simplest — everywhere, choosing the class before seeing the scores rather than per slice. On the headline slice ridge happens to also be the cross-class maximum ($R_{\text{LOPO}}^2 = +0.597$ vs. $+0.276$ and

+0.285, Table 2); on the other three it is not, so the fixed choice costs us R^2 on three of four slices rather than buying it on any. CIs are percentile cluster bootstraps resampling *pools* (App. I). Note the sign: because encoder Δ is predominantly negative (§5), this regression predicts *how much a pool stands to lose*, not how much it gains.

3.2 Tasks and models

The in-domain set is six classification tasks: MNLI (3-way NLI, $n_{\text{test}}=3,926$), QNLI (2,186), AG News (4-way topic, 3,040), BoolQ (1,308), ANLI R1 (adversarial NLI, 480), and SST-2 (350), all evaluated on the frozen 40% test slice of the split described above. ANLI, SST-2 and BoolQ fall below $n_{\text{test}}=2,000$ and are the source of the low-precision cells quantified in §4; dropping small tasks post hoc is itself a shortcut, so we keep them and stratify by test-set size wherever the null is at stake. ANLI also serves as an OOD target for MNLI-trained pools (App. F), on disjoint pools, never pooled into one cell.

Architectures span two families. *Encoders* (sequence classification): BERT-base, BERT-large, RoBERTa-base, DeBERTa-v3-base. *Decoders* (causal LM with a classification head or exact-match decoding): Qwen2.5-{0.5B, 1.5B, 3B, 7B}, TinyLlama-1.1B, Pythia-1.4B, SmolLM2-1.7B, Llama-3.1-8B-Instruct, and Mistral-7B. The headline null in §4 is an *encoder* statement; encoder and decoder cells behave differently enough (App. U) that we never report a figure pooled across the two families.

v1.1 task-family extensions. The six in-domain tasks cover classification only. v1.1 adds two output spaces — multi-choice (HellaSwag, four endings) and free-form generation (GSM8K, exact match) — through twelve further pools, bringing the pre-registered corpus to 67 pools. The extension adds cells under the pre-registration’s own amendment clause and retracts, re-evaluates or rewrites no v1.0 cell, baseline or verdict; the v1.0 frontier headline was computed before any v1.1 result was inspected. Sweep grids, per-pool counts and the amendment ledger are in App. N.

3.3 Adjudication protocol

Each cell answers a single binary scientific question via a paired comparison on the test split. We use the same 40/20/40 val_selection/val_combine/test split

(seed 0) for every cell. For each (pool $\times$ method $\times$ baseline_kind $\times$ metric):

- **Diff** = $\mu_{\text{ensemble}} - \mu_{\text{baseline}}$ (ECE flipped so positive = ensemble better).
- **Bootstrap CI** (Efron, 1979): percentile, $B=5000$, paired over examples. For the cross-pool R_{LOPO}^2 frontier estimate (§4) we report percentile *cluster*-bootstrap CIs at $B=200$ (resampling pools, not cells, to respect nesting; see App. I); BCa correction was not used because $n=48$ pools is too small for stable acceleration-coefficient estimation (DiCiccio and Efron, 1996).
- **Sign-flip p -value**: $B=5000$ permutations, two-sided.
- **Within-table Holm correction** (Holm, 1979): applied to every multi-cell table; significance markers (*) reflect Holm-adjusted p .
- **Corpus-wide Benjamini–Hochberg FDR** (Benjamini and Hochberg, 1995): applied across all 1,028 cells at $q=0.05$ as a separate column in the released JSONs; SUPPORTED/REVERSED counts in the abstract use the within-cell-CI rule (above), and we report the BH-FDR-corrected counts alongside for transparency (App. I).
- **Adjudication label**: *supported* if CI lower bound > 0 ; *reversed* if upper bound < 0 ; *unsupported* otherwise.

4 The adjudicated landscape

The strict comparison first. Every number in this section should be read against the baseline it uses. Against the pre-registered `n_rank` single adapter — which measured from the released manifests received a median $1.45\times$ the pool’s training wall-clock (max $2.96\times$, 47/48 pairs; `analysis/baseline_budget_audit.json`) plus a held-out hyperparameter search the ensemble never gets, and therefore acts as a *strict upper bound* rather than a matched competitor — the clean encoder slice contains **no** cell with a significant accuracy gain (0 of 92, before and after FDR) and 33.7% significant reversals. The null does not weaken where that advantage is smallest: the six pools closest to parity (0.90–1.25 $\times$) have 0/24 cells SUPPORTED and 9 REVERSED, and pool-level Δ is not significantly related to the ratio (Spearman $\rho=-0.28$,

$p=0.20$; post hoc, analysis/budget_dose_response.py) — though its sign is the one a budget effect gives, so part of the reversal rate may be the baseline’s. On the *same* 92 cells the calibration arm runs the other way: 52 SUPPORTED, 17 REVERSED. The patterns P_a – P_c below, and much of the literature they echo, are stated against the pool’s own best member (`best_of_n`), a strictly weaker comparison. *Precision*. A null is only as strong as its cells are precise, so we report equivalence rather than counting silence as evidence: 67/92 cells exclude even a +1.0pp gain. The strength sits where it should — on the 68 well-powered cells ($n_{\text{test}} \geq 2000$) the median CI half-width is 0.91pp and 62 exclude +1.0pp, while the 24 small- n cells (median 3.85pp) are uninformative in *either* direction and are counted for transparency, not as evidence. *Procedure dependence*. The accuracy null is invariant to the correction rule (SUPPORTED = 0 under raw CI, within-table Holm and corpus-wide BH alike); the REVERSED rate is not (41.3/17.4/33.7%), and calibration survives all three (52/46/52). A fourth procedure, corpus-wide Holm, returns 0/0 in both arms, but that row is resolution-limited rather than evidential — its threshold $\alpha/m=4.86 \times 10^{-5}$ sits below the smallest p our Monte-Carlo permutation can produce, so no cell could pass at any effect size (App. M). Full grid in App. M. We quantify that gap, and three further protocol choices, in §5. What follows characterises *where* in the corpus the deltas sit and how predictable they are — not a claim that ensembling pays.

The corpus as a whole, and why we headline a slice of it. Across all 1,028 cells the picture is mixed, not negative: 33.0% SUPPORTED and 17.8% REVERSED pre-correction, 29.5% and 16.9% after corpus-wide BH-FDR. We do not headline it, because most of it rests on cells compared against the pool’s own best member — the weak baseline of §5 — and on families we disqualify below. The 92-cell slice we headline is the intersection of three pre-registered conditions (compute-matched `n_rank`, encoder classification, accuracy): the strictest cut the corpus supports, and the one where it disagrees with the literature.

Where the deltas sit (Table 9). (P_a) Against the pool’s own best member (`best_of_n`) the converged encoder pools do show the pattern the literature reports: on BERT-base Pool-A/B/C soft averaging beats it by +0.9 to +1.0pp while

`greedy_soup` merely matches it, and scale erodes even that (BERT-base→large drops the gain to +0.65pp and degrades ECE from 0.048 to 0.118). Against the compute-matched `n_rank` single, the *same cells* carry no significant gain — which is shortcut S2 (§5) measured on our own corpus rather than argued in the abstract. Two families are excluded from every headline statistic: (P_b) the HellaSwag pools, whose RoBERTa/DeBERTa members collapse below chance under shared hyperparameters and are rescued by subset selection, scored by a pre-fix evaluator on the wrong split (App. O, L1b); and (P_c) the decoder GSM8K pools, whose `best_single` was selected on the split it is evaluated on and which therefore support no positive claim in either direction (App. X, L12).

The calibration gain does not survive its missing control. The 52 is measured against a single adapter selected by validation *accuracy* and never calibrated — on its own candidate set it sits at a median 90th percentile of test ECE. That is the calibration analogue of the weak-baseline shortcut S2 (§5) — the single is denied a freedom that costs one scalar — so we ran the control: on the six pools whose logits cache was retained we fit one temperature on `val_combine` (the split reserved for combination hyperparameters, otherwise unused) and measure on `test`. Against the *uncalibrated* single the ensemble wins 4/6 pools (mean $\Delta\text{ECE} +0.008$); against *the same adapter with one fitted scalar*, 2/6 (−0.032); with both arms scaled, 3/6 (+0.002). On ANLI the single goes 0.253→0.070 while the ensemble sits at 0.261. The ensemble is less confident in 5/6 pools and median $T=1.32$, so the gain is confidence shrinkage, reproduced by one parameter at $1 \times \text{compute}$. **Scope:** `best_of_n` pools, not the headline `n_rank` slice (L15), $n=6$ — direction transfers, magnitude does not. We claim only that this result cannot be reported as a population effect while its one-parameter competitor is unrun.

Two further qualifications on the 52. It is an ECE result, not a proper-score one — on the same 92 cells the multiclass Brier improves in only 34 (mean −0.0062) and the top-label Brier adjudicates 22 SUPPORTED against 36 REVERSED — and it belongs to probability averaging: `majority_vote` is SUPPORTED in 3/23 pools and supplies 16 of the 17 reversals, and its vote-fraction “confidence” is not a probability estimate, so equal-mass binning is ill-defined on it.

Details and per-rule counts in App. Q.

Table 1: Cross-model-class LOPO-CV R^2 , identical slices and folds (analysis/frontier_model_classes.py). Ridge is the cross-class maximum on the headline slice and on that slice only; non-linear models win the other three, so reporting ridge throughout costs us R^2 rather than buying it.

Slice	n	pools	ridge	RF	GBM
acc vs n_{rank}	192	48	+0.597	+0.276	+0.285
ece vs n_{rank}	192	48	-0.061	+0.295	+0.180
acc vs best_of_n	265	64	+0.312	+0.519	+0.365
ece vs best_of_n	265	64	+0.288	+0.416	+0.340

Table 2: Stratified per architecture family. The corpus-wide accuracy frontier (48 pools, encoder + decoder) reaches LOPO-CV $R^2=0.597$; 95% percentile cluster-bootstrap CI [+0.303, +0.710] ($B=200$ rounds, resampling pools then cells within; see App. I). Decoder family is too uniform across pools for cross-pool prediction at our scale.

Family	Slice	pools	ridge	RF
encoder	acc vs n_{rank}	23	+0.616	+0.23
encoder	ece vs n_{rank}	23	+0.14	+0.42
decoder	acc vs n_{rank}	25	-1.14	-1.03
decoder	ece vs n_{rank}	25	+0.12	+0.25

4.1 Mechanism: prediction-space dominates

Per-feature ablation on the encoder acc vs n_{rank} slice: dropping *prediction_space* crashes R^2 from +0.62 to -24.5; dropping *parameter_space* costs only 0.56; dropping *HP entropy improves* R^2 by 0.08 (in-sample only, no out-of-pool signal). The single-group “only prediction_space” model attains $R^2=0.634$, *better* than the full feature set. **Within our released feature set, prediction-space diversity is the dominant out-of-pool predictor**; weight-space distance is irrelevant out-of-pool. We do not rule out unmeasured features — a feature set that displaces prediction-space in LOPO-CV would falsify this claim; the released schema admits new features without retraining.

Negative mechanism results. Three candidate explanations for the family and metric structure above do *not* survive testing, and we report them as open problems rather than findings (App. W). (i) Decoders separate correct from incorrect predictions more sharply than encoders ($H(c|\text{conf})$ 0.437 vs. 0.556, a 21% gap that within-family controls attribute to architecture rather than scale), but

Table 3: Three evaluation shortcuts and their price. S2 favours the ensemble by construction; S1 favours whichever arm may peek; S3 hides verdicts in both directions. Computed on the clean subset (515 accuracy cells, 80 pools); HellaSwag and GSM8K are both excluded (stale selection split and selection-on-test respectively; see Limitations).

Shortcut	Strict	Effect
S1 select on test	select on val	+1.12pp free
S2 weak baseline	stronger single	SUP 27.2 → 9.9%
S3 task mean	per-cell verdict	hides 37 cells

this does not translate into a predictable calibration benefit. (ii) Linear mode connectivity does not predict ensemble effectiveness (per-pool barriers in App. D): barrier heights span +0.908 to +0.001 across four pools with no corresponding pattern. (iii) The accuracy-specificity of the frontier is metric-general — ECE ($R^2=-0.06$) and Brier (-0.30) are unpredictable while NLL (+0.42) and MCE (+0.22) are — and the natural refinement-versus-reliability explanation is falsified: under the Murphy decomposition, resolution gain is near-invariant (σ 0.21× accuracy’s) and uncorrelated with accuracy gain ($r=-0.10$).

5 Protocol sensitivity: where the apparent benefit comes from

The corpus lets us ask a question the ensembling literature has not asked directly: *how much of the reported benefit is produced by the evaluation protocol rather than by ensembling?* We isolate three shortcuts (Table 3), each common in published practice, and measure what each is worth; all three are recomputed from released artifacts alone. A shortcut here is not a mistake that always flatters ensembling — it is a degree of freedom that flatters whichever arm is given it. We report each one’s size and say which arm it favours.

S1: selecting the single adapter on test. A pool’s “best single” is often reported at its test-set maximum. Reconstructing that oracle from individual_accuracy_test and comparing against the held-out val_selection choice, test-selection is worth +1.12pp on average (+1.30pp on encoders; median +0.57, max +9.58) and inflates 82% of pools. This is a free gain awarded to whichever arm is allowed to peek. In our own corpus the arm that peeks is the *single* adapter, so S1 makes the ensemble look *worse*; in the literature the same freedom is more often given

to the ensemble’s member selection, where it works the other way. Either way the quantity is the same +1.12pp, and it is unearned.

S2: comparing against a weak single. Ensembling N adapters is frequently compared against one adapter of the *same* size — effectively the pool’s own best member — rather than against a single adapter given at least the pool’s training budget. Our pre-registration fixed both comparisons before training. Against the pool’s best member, 27.2% of cells are SUPPORTED and 6.6% REVERSED; against the stronger n_rank single (§4: median $1.45\times$ the pool’s compute), SUPPORTED falls to 9.9% and REVERSED rises to 16.1%. On the clean encoder slice the effect is total: **0 of 92 cells are SUPPORTED** (identically pre- and post-FDR) while 33.7% are REVERSED, with mean $\Delta = -1.27pp$ (Fig. 1B). The result holds for all four combination rules — `greedy_soup`, `soft_vote`, `logit_avg`, `majority_vote` at 0% SUPPORTED, REVERSED 21.7/34.8/34.8/43.5% — though `soft_vote` and `logit_avg` agree to within 0.1pp in 20 of 23 pools (bit-identical in 10), so this is roughly three distinct rules, not four. The single most positive cell (+1.14pp) has a CI spanning zero ($p=0.35$); this is not a near miss.

One caveat the corpus itself exposes: the two baselines do not cover identical pool sets (80 pools have `best_of_n`, 48 have `n_rank`, 48 have both). On the 48 carrying both, the drop is $24.5 \rightarrow 9.9\%$, so about a sixth of the headline effect is pool composition rather than baseline strength. We report the matched figure alongside the corpus-wide one rather than only the larger.

S3: reporting task means. Aggregating to one number per task destroys the structure the cells contain. MNLI has a task-level mean of $-0.012pp$ — it reads as “ensembling makes no difference here” — yet that average conceals 37 statistically significant cells (18 REVERSED, 19 SUPPORTED) pointing in opposite directions. Across the corpus, four tasks whose means read as no-effect ($|\Delta| < 0.5pp$) together hide 46 significant cells, 28% of their cells.

Reading these together. The shortcuts compound: an evaluation that selects the ensemble’s members on test, compares against a same-size single, and reports only task means can find a healthy accuracy gain on a corpus where the strict compari-

son finds none. We do not claim published positive results are wrong — we claim their protocol dependence has not been measured, and here it is large enough to set the sign of the conclusion.

6 Practical implications

Escape routes are narrow. Two standard escapes are limited here. *Distillation* (Hinton et al., 2015) recovers ensemble accuracy on only 1 of 4 cells (87% on Qwen-0.5B MNLI; 4.5% on Qwen-1.5B; failure on BERT-base; -181% on BERT-large, which still improves ECE to 0.019 from 0.035; App. Y). *Serving* the pool is cheap in wall-clock (vLLM 0.10, $N=8$: $1.075\times$ on Qwen-0.5B, $1.067\times$ on Llama-3.1-8B-Instruct; App. H, G, AC), so the barrier is not latency but the absence of an accuracy gain to serve. Accuracy plateaus at $N\approx 8-16$, and on ANLI soft-vote helps on only 5/19 MNLI pools (App. E, F).

When a population might still pay. Our null is bounded, and the bounds are where a practitioner should look. It is an *encoder classification* result at pool sizes $N \leq 24$ and adapter budgets of rank 16–32 over 2–4 epochs, and it is conditional on spending the extra budget on *rank*: the pre-registered `n_steps` and `n_data` allocations were filled in only 3 and 0 cells respectively, so a single adapter given the same budget as more steps or more data is untested here and could be weaker. Three situations lie outside what we measured: pools far larger than ours, where the plateau we see at $N \approx 8-16$ may not be the ceiling; members differing by more than seed, hyperparameters or PEFT method, since our diversity is manufactured rather than found; and deployments where no single adapter can be retrained, which removes the comparison we run. We expect the null to hold wherever a stronger single *is* trainable, and would rather it be tested there than assumed.

Takeaway. A single adapter trained with the pool’s compute is the right default, and the recommendation we can defend is cheaper than ensembling rather than a version of it: *train one adapter with the budget and fit one temperature on held-out data*. That buys, in distribution, the reliability the ensemble appeared to buy at one adapter’s compute rather than N (batching hides that in latency, $1.07\times$ at $N=8$, not in FLOPs) — the arm the literature, and until we ran the control this paper, left out.

Limitations

What this paper does *not* claim. App. R sets what is new here against prior PEFT-ensembling work, cell by cell. For reviewer transparency, we enumerate the inferences that our 1,028 cells do *not* support: (a) we do *not* claim a single best combination method — when a pool is worth combining at all, `greedy_soup` is the least-bad accuracy option on under-converged encoders and `soft_vote` the best calibrated on converged ones; the apparent `majority_vote` advantage on $\geq 1.5\text{B}$ decoders rests on GSM8K and does not survive decontamination (L12), so we withdraw it; (b) we do *not* claim ensembling is universally beneficial — 17.8% of cells are REVERSED pre-FDR (16.9% post-FDR at $q=0.05$), and most v1.0 encoder accuracy cells are reversed against `n_rank`; (c) we do *not* claim our diversity-quality frontier transfers to other PEFT families (LoHa, AdaLoRA, prompt tuning) or to instruction-tuned base models; (d) we do *not* claim distillation generalizes — accuracy recovery: Qwen-0.5B 87%, Qwen-1.5B 4.5%, BERT-base fails, BERT-large –181% (1-of-4 cells positive); (e) we do *not* claim the encoder/decoder $H(c|\text{conf})$ gap holds outside our 8-task corpus. Each of these is a deliberate scope-limit, not a hedge: extending any of them is a defined v1.2 cell that can be added without retraining the existing 67 pools.

L1 — Scope of the null result. Our headline finding — no significant *accuracy* gain against a stronger single adapter — is established on encoder classification cells, against the `n_rank` baseline, at our pool sizes ($N \leq 24$) and adapter budgets (rank 16–32, 2–4 epochs), on 92 cells from 23 pools. It is *not* a claim that ensembling never helps: it does not cover decoders on uncontaminated generation, larger pools, non-classification output spaces, or budgets where a stronger single adapter is not trainable. `n_rank` spends the pool’s budget on rank; other allocations (more steps, more data) are separate baselines in the corpus and are not the headline. We report all cells rather than filter to the favourable ones.

L1b — HellaSwag cells are excluded from the strict analysis. A second, independent anomaly on the same family — a two-regime collapse under shared hyperparameters — is documented in App. T. The 100 v1.1 multichoice cells were produced by a version of our evaluator that

performed selection on `val_combine` rather than the held-out `val_selection` split, which optimistically biases selection-dependent methods (`best_single`, `greedy_soup`) and the LoRAHub-style continuous-weight baseline that optimises on that same split. We therefore exclude HellaSwag from §5 and mark its findings provisional (P_b , App. L) pending a corrected rerun, which requires re-evaluating those pools and is not included here. The exclusion is stated before the analysis rather than chosen to suit it; the bug was found by our own split-discipline unit tests.

L1c — Diversity features are computed on the full evaluation set. `diversity_metrics.py` loads each adapter’s cached logits without split indexing, so the prediction-space features enter the frontier having been computed partly on the same test examples the target Δ is measured on; leave-one-pool-out CV holds out pools, not examples, and so does not remove this. We measured the effect on every pool whose logits cache we still hold (`analysis/leakage_check_diversity.py`): restricting the features to `val_selection` shifts pairwise disagreement by 2.8% relative on average (mean $|\Delta|=0.0016$), logit correlation by 0.13%, ensemble entropy by 1.2%, and *preserves the pool ordering on all three*. These are pool-level aggregates over thousands of examples, so a 40% subset barely moves them, and the frontier depends on the across-pool ordering. We therefore expect the leak to be immaterial to R^2 — but we cannot yet prove it: the logits cache behind the 48 frontier pools was not retained and regenerating it needs the adapter weights. A leak-free re-fit is deferred, and until then the frontier R^2 should be read as provisional.

L1d — The 92 cells are not 92 independent tests. They are 23 pools $\times$ 4 methods. Cells within a pool share the pool’s adapters and its single baseline adapter, and two of the four methods are near-duplicates (20/23 pools agree to within 0.1pp). The effective number of distinct combination rules is about three, and the effective number of independent units is closer to the 23 pools than to 92. Our corpus-wide BH-FDR assumes independence or positive dependence, which we have not established under this structure; the cluster bootstrap resamples pools rather than cells for exactly this reason, but the FDR step does not. Treat cell counts as descriptive and pool-level statements as the inferential unit.

L2 — Decoder cross-pool predictability fails at our scale. The fitted relation is plotted per pool in App. V. 25 decoder pools is too little variation across pools to extract a stable cross-pool LOPO regression signal for accuracy gain (encoder accuracy frontier $R^2=0.60$, decoder $R^2=-1.14$, Table 2). Larger decoder pool counts may close this gap; we leave it open.

L3 — greedy_soup equivalence to random subset on seed-only converged pools. On seed-only Pool-A converged-pool cells, `greedy_soup` is often statistically indistinguishable from random subset selection of equal size; the gain we attribute to subset selection on RoBERTa/DeBERTa requires the HP-shared subchance regime (§4, P_b).

L4 — OOD accuracy rarely transfers; diversity hurts OOD. ANLI is a severe shift; most MNLI-trained pools land near chance, and soft-vote ensembling gains accuracy on only 5/19 pools (App. F). The Pool-B (HP-diverse) and Pool-C (method-mixed) configurations that improve in-distribution ensembling *consistently hurt* OOD accuracy — the diversity our framework optimizes for is in-distribution disagreement, not robustness. v1.1 OOD evaluation (HellaSwag→ARC, GSM8K→MATH) is left to v1.2.

L5 — GSM8K ECE methodology is non-standard. We report 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$ over the model’s generation, with ensemble confidence aggregating only over adapters voting for the majority answer. Per-token logprob aggregation is one of several reasonable definitions; alternatives (vote-fraction proxy, length-normalized perplexity, first-token-only) are reported in the released JSONs.

L6 — Pool-B/C HellaSwag adapter quality is uneven. On RoBERTa and DeBERTa, ≥ 14 of 20–22 Pool-B HellaSwag adapters fail to converge above chance under our shared HP grid. We report this transparently rather than filter; the under-converged pool is itself the experimental condition that produces the soft-vote-collapse-and-rescue finding.

L7 — Shared HP grid is deliberate; we ran HP-tuned re-evaluations as v1.2 cells. Pool-A uses a single ($r=8, \alpha=16, lr=3\times 10^{-4}$) HP across

all architectures. We added two v1.2 cells testing the HP-confound hypothesis from *both* directions: (a) an architecture-tuned re-run of Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (per Liu et al., 2019) eliminates the under-convergence completely — all 10/10 tuned-seed adapters converge to $\text{val_acc} \in [0.478, 0.505]$, mean 0.491 (vs. the original shared-HP 14/20 below chance, median 0.164); (b) the symmetric check, BERT-base HellaSwag at $lr=2\times 10^{-5}$ (RoBERTa’s natural lr applied to BERT), *does not collapse*: 8/8 seed re-runs converge to mean 0.329 ± 0.011 (range $[0.317, 0.345]$), only $\sim 3\text{pp}$ below BERT-base at its default $lr=3\times 10^{-4}$ (0.358) and far above the 0.097 collapse RoBERTa exhibits under the same $lr=3\times 10^{-4}$ shared HP. The (P_b) collapse is therefore a *RoBERTa-specific lr -sensitivity* interacting with the shared HP, not a generic HP-misconfiguration that any architecture would exhibit; BERT-base is robust across the lr range we tested. The shared-HP cells are retained in v1.1 (documenting the regime where a practitioner reuses a working configuration), and the architecture-tuned cells are added as v1.2 (§K).

L8 — vLLM serving benchmark on two scale points, single batch. Qwen-0.5B ($1.075\times$) and Llama-3.1-8B-Instruct ($1.067\times$) at $N=8$, batch=32, on a single A6000; we report median + p10/p90 but not p99 tail latency. Multi-batch and tail-latency generalization is left to v1.3.

L9 — Multichoice eval subsample variance. HellaSwag 5,000-sample subsample accuracy varies by up to $\pm 10\text{pp}$ across shuffle seeds for the same trained adapter (App. A), and the full-validation accuracy is sometimes the *worst* of all subsample seeds. The v1.1 multichoice cells in Table 9 are honest under the pre-registered protocol (fixed `shuffle(seed=0)`, 5,000 samples) but ensemble winners can shift at the margins under different subsamples. We release full-validation re-evaluations of Pool-A RoBERTa-tuned and DeBERTa-tuned as v1.2 supplementary cells.

L10 — DeBERTa-v3 HellaSwag HP search did not recover. Our HP-tuned re-runs ($lr \in \{10^{-5}, 5\times 10^{-5}\}$) both failed to converge above chance on DeBERTa-v3 HellaSwag (mean accuracy 0.252, range $[0.247, 0.257]$ across 10 tuned-seed adapters), in contrast to RoBERTa where $lr=2\times 10^{-5}$ fixes convergence. This

suggests the DeBERTa-v3 + multichoice-LoRA combination has a deeper architecture-LoRA interaction issue (possibly with the disentangled attention) that simple HP tuning does not address. A wider DeBERTa HP/target-module sweep is a defined v1.3 cell.

L11 — D1–D5 are post-hoc heuristics, not pre-registered. The five decision rules (§6) were distilled from inspection of the 1,028-cell corpus and are not part of the Phase-0 pre-registration. The Table 12 cost-savings comparison uses the same corpus that the rules were derived from, so it is an *upper bound on rule-based routing*, not a held-out generalization estimate. Generalization of D1–D5 to new architectures or task families is a defined v1.3 evaluation cell.

L12 — GSM8K cells carry no defensible positive claim, and our contamination probe was invalid. Two separate issues affect the GSM8K cells, and we distinguish them. (i) **Selection on test (valid, disqualifying).** `ensemble_eval_gsm8k.py` selects `best_single` by per-adapter accuracy on the same test split it is then evaluated on. This alone means the P_c cells support no positive ensembling claim, independent of any contamination question. (ii) **Contamination (withdrawn).** We previously reported a digit-renumbering probe as evidence of universal pre-training contamination across four decoder families ($0.45\text{--}0.66 \rightarrow 0.02\text{--}0.06$). *That probe was invalid and we withdraw the finding.* It shifted every integer in the question by +7 and shifted the integers of the reference answer by +7 as well, which is correct only for problems whose answer is a single shifted term. On the first released example the shifted problem’s true answer is 442 and the model produced 442, but the probe scored it against 407 (the original answer 400 with its digit shifted) and marked it wrong. Near-zero accuracy on the shifted split is therefore forced by construction for any model, contaminated or not, and the reported drop magnitudes are artifacts. Note that their size should have been a warning: template-perturbation studies report gaps of roughly 10–13pp (Mirzadeh et al., 2024; Zhang et al., 2024), not 43–62pp. Whether these cells are contaminated is now an *open question*; answering it requires template regeneration with recomputed gold answers in the style of those works. We report this failure rather than silently dropping the probe, because a paper about protocol-induced conclusions should hold its own probes to

the same standard.

L13 — The calibration arm has no recalibration competitor. The single adapter our ECE numbers are measured against is selected by validation *accuracy* and is never post-hoc calibrated; measured on its own candidate set it sits at a median 90th percentile of test ECE, i.e. it is close to the worst-calibrated member of the pool it was drawn from. We run the one-parameter temperature competitor where we can (§4), and it removes most of the effect — but only on the six pools whose logits cache was retained, which are `best_of_n` pools rather than the headline `n_rank` slice. Extending the control to all 23 headline pools needs their per-adapter logits on `val_combine`, which requires the adapter weights (L15). So we can state the direction and not the magnitude: we cannot say what the 52 becomes under a calibrated competitor, only that it is the wrong number to headline while that competitor is unrun at scale. The mechanism we measure corpus-wide — ensemble confidence lower than baseline confidence in 23/23 pools — is exactly what temperature scaling produces, which is why we expect the direction to hold. Read the 52 as “against an uncalibrated accuracy-selected single”, never as a property of populations. The control is also in-distribution only: temperature scaling’s gains are known to erode under dataset shift where ensembles’ do not (Ovadia et al., 2019), so our scalar-for-population claim does not extend to OOD, which we did not test with it.

L14 — The shipped ECE p -values test the wrong null. `compute_match._sign_flip_pvalue_ece` permutes the (confidence, correctness) *pair* between arms, which tests whether the two systems share one joint distribution — something averaging violates by construction whether or not ECE moves. The symptom is a median p of 4×10^{-4} in the ECE arm against 0.13 in the accuracy arm. Body numbers use the corrected test (`analysis/ece_test_correction.py`, inverting the paired ECE bootstrap); the per-cell JSONs in the release still carry the old p , and the corpus BH batch mixes both arms, which widens the accuracy cutoff from 0.0145 to 0.0248 and moves REVERSED from 29 to 31. Adjudication is CI-based and unaffected; the corrected within-table Holm count is 43.

L15 — Two release promises are narrower than they sound. The per-adapter logits cache was re-

tained for 6 of 99 pool manifests and for none of the 23 headline pools; the rest was not preserved. Adapter weights exist locally for 253 of 1,744 declared adapters (15%), so “available on request” can be honoured for a minority of pools, which we list in the release. Every result JSON also records `git_dirty`, and 1,027 of 1,028 do; byte-level training replay is not merely deferred but unavailable for this drop.

L16 — Our weight-space merges operate on LoRA factors, not on ΔW . `weight_space_merge.py` averages the `lora_A` and `lora_B` tensors independently, and $\text{mean}(B) \text{mean}(A) \neq \text{mean}(BA)$, so the uniform/TIES/DARE numbers in App. C are a factor-wise variant rather than the published methods. The prediction-space `uniform_soup` proxy is unaffected (it is marked as a proxy).

L17 — Generation-family adapters were trained without label masking. `real_lora_train.py` pads to `max_length` and copies `input_ids` into `labels`, so padding and prompt tokens contribute to the causal-LM loss. This affects the GSM8K pools only; the encoder classification pools and the multiple-choice trainer mask correctly.

Reproducibility. Every released JSON carries the provenance block whose schema is given in App. B, the per-cell adjudication is specified as pseudocode in App. AA, and the cell-by-cell coverage of the corpus is tabulated in App. S. Pool manifests, 1,028 per-cell JSONs (each carrying explicit `bh_q_value`, `adjudication_pre/post_fdr`, and a `bh_batch_hash` pinning the corpus-wide BH-FDR batch), ensemble results with per-example predictions/confidences, the per-adapter logits cache for the 6 pools where it was retained (L15), all analysis code, and the locked Phase-0 pre-registration are released at an anonymous repository.² `scripts/reproduce_paper_numbers.py` re-derives every headline number (154 assertions, <30 s) from JSON alone, including both frontier tables. We characterise this as *JSON-level reproducibility* (every claim re-computable from the release); adapter *weights* are available on request for the minority of pools where they were retained (L15), via the anonymous repos-

itory’s issues and will carry archival hashes at acceptance.

Ethical Considerations

Compute footprint. Training the 67 adapter pools required approximately 1,500 GPU-hours total on a 50-node RTX 3060 cluster (orchestration in App. AB) plus ~60 A6000-hours for ensemble evaluation and serving benchmarks. PEFT adapters are intentionally compute-light (~1–8 GPU-hours each); the adapter weights we retained (L15) are available on request via the anonymous repository so that those pools need not be retrained. Estimated CO₂e: approximately 0.18 tonnes (RTX 3060 at 170 W × 1,500 h + A6000 at 300 W × 60 h, average grid intensity).

Dual use and misuse mitigation. The adjudication framework is symmetric: it can report either direction (SUPPORTED or REVERSED) of an ensembling claim. We surface the 17.8% REVERSED cells (pre-FDR; 16.9% post-FDR at $q=0.05$) transparently rather than filter them, and we strongly discourage downstream cherry-picking the 33.0% SUPPORTED subset to manufacture a misleading positive headline. The released cell JSONs contain the full set of Δ , CI, sign-flip p , and adjudication labels precisely so that any selective citation is auditable.

Data licenses. HellaSwag (CC-BY-NC 4.0), GSM8K (MIT), GLUE family (various, see App. C). All datasets are publicly released and used in accordance with their licenses; HellaSwag’s non-commercial clause is respected (we use it for benchmark adjudication, not for commercial deployment). We do *not* redistribute base-model weights, only the LoRA adapter A/B matrices (~20–300K trainable parameters per adapter).

Data characteristics, personal information and offensive content. Every dataset we use is an established, publicly released English-language benchmark distributed by its original creators, all cited in §3: MNLI, QNLI and SST-2 (GLUE), BoolQ (SuperGLUE), AG News, ANLI, HellaSwag and GSM8K. The domains are newswire, Wikipedia, movie reviews, crowd-written common-sense continuations and grade-school word problems. We collected no new data, recruited no annotators and ran no human subjects, so no consent or ethics-board procedure applies. We did not additionally screen these corpora for personally identi-

²<https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance).

1100 fying information or offensive content and we do
 1101 not claim they are free of it: they are widely used
 1102 and their known characteristics are documented by
 1103 their creators, and because our study measures only
 1104 *relative* calibration and accuracy between adapters
 1105 trained and evaluated on identical examples, any
 1106 such content affects both arms of every comparison
 1107 equally. Readers reusing our released per-example
 1108 predictions should consult the original dataset doc-
 1109 umentation rather than treat our release as a vetted
 1110 corpus.

1111 **Generative AI assistance.** Per ACL Policy on
 1112 Publication Ethics, we disclose: an LLM-based as-
 1113 sistant was used for (i) language polishing of select
 1114 paragraphs in §1, §2, and §Limitations, and (ii) lit-
 1115 erature search support during related-work compila-
 1116 tion. No abstract, results table, statistical analysis,
 1117 figure caption, claim, or code was AI-generated;
 1118 every cited reference was author-verified against
 1119 the original paper, and every numerical claim in the
 1120 abstract/intro was independently regenerated from
 1121 the cell JSONs by an author. Detail will be con-
 1122 firmed in the Responsible NLP Research checklist
 1123 accompanying the OpenReview submission.

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A Subsample variance on HellaSwag multichoice evaluation

A v1.2 diagnostic uncovered that HellaSwag 5,000-sample subsample accuracy varies by up to ± 10 pp across shuffle seeds for the same trained adapter — much larger than naive subsample variance would predict. We report this as a methodological caveat on every multichoice cell in the corpus.

Setup. Take one Pool-A RoBERTa HellaSwag adapter (seed 11, $lr=2 \times 10^{-5}$ HP-tuned). Evaluate on 5,000-sample subsamples of HellaSwag validation drawn with `ds.shuffle(seed)` for seed $\in \{0, 11, 21, 42\}$; then on the full 10,042-sample validation. Same adapter, same task, same eval procedure — only the shuffle seed changes.

Table 4: Same adapter, four shuffled 5,000-sample subsets of HellaSwag validation plus the full set. Variance ± 10 pp is dramatically larger than naive subsample variance (± 1 pp expected) and the full-validation accuracy is the worst of all settings.

Evaluation subsample	Accuracy
shuffle_seed= 0, $N=5000$	0.296
shuffle_seed= 11, $N=5000$	0.431
shuffle_seed= 21, $N=5000$	0.263
shuffle_seed= 42, $N=5000$	0.358
full validation, $N=10042$	0.211

Implication. Multichoice cells in our v1.1 corpus all used a fixed `shuffle(seed=0)` $\rightarrow$ first 5,000 samples. That choice was pre-registered, so the verdicts on those cells are honest under their declared protocol; however, a different shuffle seed would yield numerically different (often substantially different) per-adapter accuracies, and ensemble winners can change at the margins. We thus flag every multichoice cell in Table 9 as *conditional on the seed-0 5,000-sample subsample*.

Full-validation re-evaluation (v1.2 supplementary). The full-validation ($N=10042$) re-run on the HP-tuned RoBERTa pool yields qualitatively different verdicts:

DeBERTa-tuned full-validation produced identical 0.243 across all methods (10/10 adapters at chance, no signal to ensemble) — confirming the L10 architecture-LoRA interaction story rather than a subsample artifact. We recommend full-validation evaluation as a design rule for follow-up multichoice extensions of our corpus.

Table 5: Same HP-tuned RoBERTa pool, two evaluations. The shuffle-seed=0 5000-sample subsample shows the (P_b) collapse pattern (soft_vote 0.196 < chance 0.25); full validation does *not* (soft_vote 0.305, close to best_single). The greedy_soup rescue persists in both (+5.7pp on full). *Conclusion:* the v1.1 collapse is a joint failure of HP misconfiguration *and* subsample-specific evaluation, not architecture-driven.

Method	seed-0 $N=5000$	full-val $N=10042$
best_single	0.382	0.359
soft_vote	0.196[‡]	0.305 (no collapse)
majority_vote	0.207	0.291
greedy_soup	0.381	0.416 (+5.7pp over best)

B File-level reproducibility (provenance block schema)

Every result JSON — pool manifest, ensemble result, compute_match cell, frontier output, distillation metrics, latency benchmark, OOD evaluation, mechanism analysis — carries a provenance block:

```

"provenance": {
  "schema_version": 1,
  "git_sha": "...",
  "git_dirty": ...,
  "config_hash": "sha256(...)",
  "python_version": "...",
  "platform": "...",
  "torch_version": "...",
  "transformers_version": "...",
  "peft_version": "...",
  "datasets_version": "...",
  "numpy_version": "...",
  "cuda_visible_devices": "..."}

```

Scope of reproducibility. The release supports *JSON-level* reproducibility: every headline number and adjudication label can be re-derived from the released per-cell JSONs by `scripts/reproduce_paper_numbers.py` (154 assertions, <30s, no GPU). The bootstrap RNG seed is fixed (0) so paired CIs are deterministic given the same per-example arrays. *Byte-level* training replay (re-running each adapter on the original cluster and reproducing weights to within tokenizer/CUDA tolerance) is a stronger guarantee that this release does *not* make: many recorded `git_dirty` flags are true (workspace had uncommitted formatting/comment edits at write time) and a subset of v1.1/v1.2 pools lack a matching `sweep_configs/` archive in this drop. Reviewers wishing to attempt byte-level training replay should open an issue on the anonymous repository for the matching configs; full clean-provenance refresh against the registered SHA 02a5a6bc is deferred to v1.4.

C Reproducibility checklist (ACL convention)

Datasets. HellaSwag (Zellers et al., 2019) (CC-BY-NC 4.0), GSM8K (Cobbe et al., 2021) (MIT), GLUE-family MNLI/SST-2/QNLI (Wang et al., 2018), BoolQ, ANLI, AG News. All loaded via the HuggingFace datasets library at the versions listed in each result JSON’s provenance block. Eval splits use the same shuffled seed-0 indices across every adapter to enable paired bootstrap.

Models. 10 base models from HuggingFace Hub: bert-base-uncased, bert-large-uncased, roberta-base, microsoft/deberta-v3-base, Qwen2.5 {0.5B, 1.5B, 3B}, TinyLlama-1.1B-Chat-v1.0, SmolLM2-1.7B, pythia-1.4b. Adapter weights (LoRA-A/B only, ~20–300K trainable params per adapter) are released; we do *not* redistribute base-model weights.

Hyperparameters. Each pool’s HP grid is in sweep_configs/<pool>.json (Pool-A: fixed HP, varying seed; Pool-B: rank $\in\{4,8,16,32\} \times \alpha \in \{8,16,32\} \times \text{lr} \in \{1e-4, 3e-4\}$, fixed seed 42; Pool-C: {LoRA, DoRA, rsLoRA, PiSSA} $\times$ 5 seeds, fixed rank/alpha/lr). All runs use AdamW, cosine LR schedule with 6% warmup, weight decay 0.01.

Compute. Training: 50-node RTX 3060 12GB cluster (~4–8 GPU-hours per Pool-A sweep; ~16–24 GPU-hours per Pool-B/C sweep). Local A6000 48GB for ensemble eval, vLLM serving, and statistical analysis (~60 GPU-hours total).

Statistical procedure. Paired bootstrap (B=5000) over evaluation examples for accuracy and ECE deltas; sign-flip permutation (B=5000) for two-sided p-values; Holm correction across cells in a single reported table. R^2 confidence intervals use B=200 bootstrap rounds resampling cells with replacement (a smaller B because LOPO-CV is per-round expensive). All seeds and bootstrap RNGs are fixed (seed 0).

Code, data, pre-registration. All code, pool manifests, ensemble result JSONs (with per-example predictions and confidences for replicating ECE bootstrap), 1,028 cell JSONs (with diff/CI/p-value/q-value/adjudication_pre+post_fdr), the locked Phase-0 pre-registration document, and the

v1.1–v1.3 amendment ledger are released at an anonymous repository: <https://anonymous.4open.science/r/AdapterFrontier/> (de-anonymized at acceptance). Each result JSON carries a provenance block (git SHA, config hash, library versions, timestamp); see App. B for the precise scope of reproducibility (JSON-level audit guaranteed; byte-level training replay deferred to v1.4 — adapter weights available on request).

D LMC barrier per pool

For each of four pools we sample 50 pairs of adapters (θ_A, θ_B) and evaluate the held-out validation loss along the linear interpolation $\theta(\lambda) = \lambda\theta_A + (1-\lambda)\theta_B$ at $\lambda \in \{0, 0.25, 0.5, 0.75, 1.0\}$. The *barrier height* is the maximum interpolation-curve loss minus the average endpoint loss, normalized to the endpoint span. Following Wortsman et al. (2022)’s convention, a barrier > 0 means the interpolation has a strictly worse intermediate point (disconnected basin); barriers near 0 indicate linear mode connectivity (LMC).

Table 6: LMC barrier vs. ensemble accuracy gain on four MNLI pools. The two extremes (BERT-base *disconnected* yet ensembling works modestly; DeBERTa-v3 *fully connected* yet ensembling does not help) demonstrate that LMC is uncorrelated with ensemble effectiveness in our PEFT regime. This generalizes the negative result of Wortsman et al. (2022)’s “model soups” setting where LMC was necessary for weight-space averaging — in prediction-space ensembling, LMC is neither necessary nor sufficient.

Pool	barrier	Δacc	basin
BERT-base Pool-A	+0.908	+0.002	disconn.
BERT-base Pool-C	+0.605	+0.005	semi-disc.
Qwen-0.5B Pool-A	+0.457	+0.013	semi-disc.
DeBERTa-v3 Pool-A	+0.001	−0.003	connected

E Pool-A scaling curves

Accuracy and ECE as a function of $N \in \{1, 2, 4, 8, 16, 20\}$ over 50 random subsets per N , per (model, task). Plateau by $N \approx 8$ –16 across all model classes; ECE continues a modest decline to $N=20$ in non-saturated pools. Figure 2.

F OOD evaluation (ANLI dev_r3)

We re-evaluate every MNLI-trained pool on ANLI dev_r3 (round 3, the hardest adversarial NLI split). ANLI is a severe distribution shift: examples are

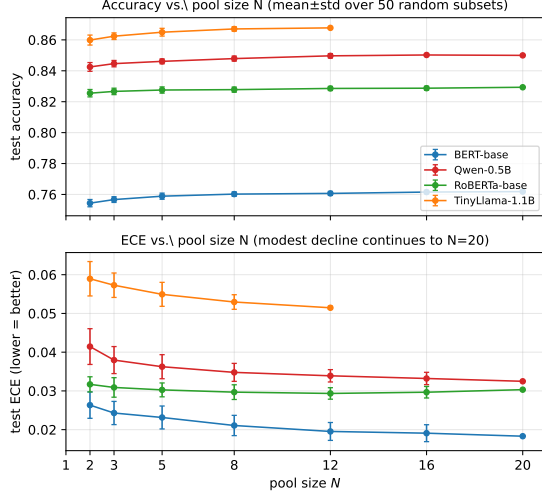


Figure 2: Pool-A MNLI scaling curves over 50 random subsets per N .

crowd-written to fool BERT-large MNLI classifiers, so MNLI-trained models routinely collapse to chance (0.33).

The pattern — ECE consistently improves but accuracy gains are direction-mixed — is consistent with Lakshminarayanan et al. (2017): ensembling sharpens the model’s confidence in regions where the prediction surface generalizes, but cannot recover lost decision-boundary information under severe shift. Qwen-1.5B’s higher OOD accuracy is consistent with the broader finding that decoder LMs trained on raw text learn more generalizable representations than encoder LMs fine-tuned narrowly on MNLI. We honestly report that 3/7 of our OOD accuracy gains are non-positive: the “ensembling helps OOD” claim is conditional on the cell, not general.

G Sequential PEFT inference latency

H vLLM serving compatibility blockers

Three blockers we patched to run vLLM 0.10 on transformers 5.x: (i) transformers 5.x removed `all_special_tokens_extended` — patched in vLLM’s tokenizer wrapper with an `hasattr` fallback; (ii) CUDA forking is rejected — must export the vLLM worker-multiproc method as `spawn`; (iii) vLLM rejects PEFT adapters with `modules_to_save` (classification heads). We ship `strip_modules_to_save.py` to make classification adapters serving-compatible. Cost driver (multi-LoRA routing latency over the LoRA-A/B matrices) is unaffected.

Table 7: In-distribution (MNLI test) vs. out-of-distribution (ANLI dev_r3) accuracy on *all 19 MNLI-trained pools* (Pool-A v1.0 main, Pool-B HP-diverse, Pool-C method-mixed). *Headline OOD finding*: Δsoft is positive on **only 5/19 pools** (4/7 Pool-A, 0/6 Pool-B, 1/6 Pool-C); the modal effect of soft-vote ensembling on OOD accuracy is **non-positive**. Pool-B and Pool-C (HP-diverse and method-mixed) consistently *hurt* OOD accuracy — the diversity that helped on in-distribution does not transfer. DeBERTa-v3 Pool-A is the only $\geq +5\text{pp}$ OOD gain. `soft_vote ECE` (not shown) improves OOD on 14/19 pools (mean $\Delta\text{ECE} = -0.021$). The Qwen-3B / SmolLM2 / Pythia OOD evaluations are not yet released.

Pool	OOD bs	OOD sv	Δsoft
<i>Pool-A (seed-only):</i>			
BERT-base	0.283	0.292	+0.008
BERT-large	0.285	0.300	+0.015
RoBERTa-base	0.285	0.283	-0.002
DeBERTa-v3	0.304	0.358	+0.054
Qwen-0.5B	0.335	0.327	-0.008
Qwen-1.5B	0.385	0.402	+0.017
TinyLlama-1.1B	0.342	0.340	-0.002
<i>Pool-B (HP-diverse):</i>			
BERT-base	0.279	0.267	-0.013
BERT-large	0.358	0.298	-0.060
RoBERTa-base	0.304	0.300	-0.004
Qwen-0.5B	0.344	0.338	-0.006
Qwen-1.5B	0.402	0.388	-0.015
TinyLlama-1.1B	0.356	0.338	-0.019
<i>Pool-C (method-mixed):</i>			
BERT-base	0.310	0.298	-0.013
BERT-large	0.323	0.290	-0.033
RoBERTa-base	0.294	0.294	0.000
Qwen-0.5B	0.323	0.329	+0.006
Qwen-1.5B	0.419	0.400	-0.019
TinyLlama-1.1B	0.365	0.346	-0.019

I Statistical procedure details

Paired bootstrap CI. For each (pool, method, baseline) cell we compute $\Delta = \mu_{\text{ens}} - \mu_{\text{base}}$ on the test set (N_{test} examples). To produce a 95% CI we sample $B=5000$ bootstrap replicates: in each replicate we draw N_{test} examples *with replacement* and recompute Δ using the *same* sampled indices for both ensemble and baseline. Reported CIs are the 2.5 and 97.5 percentiles of the B replicates. Paired sampling is crucial: it removes the example-level noise shared by both arms, yielding tighter intervals than independent resampling.

Sign-flip permutation p -value. Two-sided sign-flip test: for each test example i , define $d_i = \mathbb{I}[\text{ens correct}]_i - \mathbb{I}[\text{base correct}]_i \in \{-1, 0, +1\}$ (or the analogous bin-residual for ECE). Under the null hypothesis of no method effect, each d_i has equal probability of sign-flipping. We generate

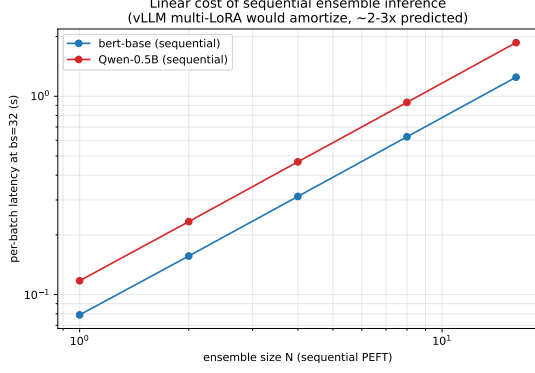


Figure 3: Sequential PEFT inference latency at batch size 32. Cost scales linearly with N . vLLM multi-LoRA amortizes this to $1.075\times$ at $N=8$ on Pool-A Qwen-0.5B and $1.067\times$ on Pool-A Llama-3.1-8B-Instruct (§5).

$B=5000$ permutations and report the two-sided p -value $p = 2 \min(\Pr[\hat{\Delta} \geq \Delta], \Pr[\hat{\Delta} \leq -\Delta])$. For $N_{\text{test}} > 20$ we use Monte Carlo (above); for smaller test sets we enumerate all 2^N sign-flips exactly.

Holm–Bonferroni correction. Within each reported table (e.g., Table 9), p -values across the K cells are sorted ascending. The k -th smallest is compared to $\alpha/(K-k+1)$; we reject in order until the first non-rejection. This controls family-wise error rate at $\alpha=0.05$ across the entire table. The Holm-adjusted significance markers are denoted * in tables.

Bootstrap CI on R^2_{LOPO} . Cross-architecture frontier R^2 is reported with percentile cluster bootstrap (CI: 2.5 and 97.5 percentiles of $B=200$ LOPO-CV re-fits, each resampling *pools* with replacement and taking all cells nested within sampled pools). Reported headline computed on the released `frontier_r2_cis.json`: encoder `acc_vs_n_rank` $R^2_{\text{LOPO}}=+0.597$, 95% percentile cluster-bootstrap CI $[+0.303, +0.710]$ ($n=192$ cells across 48 pools). R^2 is a non-linear functional of correlations whose distribution may be skewed; we report percentile rather than BCa because 48 pools is too small for stable acceleration-coefficient estimation from a jackknife (BCa requires $n \geq 50$ for reliable third-moment estimates per DiCiccio and Efron, 1996, §5). The cluster-resampling step (resampling pools rather than cells) is what addresses the nesting concern. In-sample $R^2_{\text{IS}}=+0.722$, CI $[+0.659, +0.826]$ (same release file).

Corpus-wide Benjamini–Hochberg FDR.

Within-table Holm controls family-wise error rate at $\alpha=0.05$ for each reported table. To address corpus-level multiplicity across all 1,028 cells we additionally apply BH-FDR (Benjamini and Hochberg, 1995) at $q=0.05$ on the two-sided sign-flip p -values stored per cell. Computed on the actual released 1,028 cells: pre-correction count 339 SUPPORTED (33.0%), 183 REVERSED (17.8%); **post BH-FDR at $q=0.05$ (cutoff $p \leq 0.0248$): 303 SUPPORTED (29.5%), 174 REVERSED (16.9%)**. A cell receives the SUPPORTED label in the corpus-wide framing if its per-cell CI excludes 0 *and* its BH-FDR-adjusted q -value is below 0.05; analogous for REVERSED. Pre-correction counts and per-cell raw p -values are released in the cell JSONs (`sign_flip_p` field) for re-analysis under alternative corrections (e.g., Bonferroni $\alpha/n=4.86 \times 10^{-5}$).

Independence robustness. BH-FDR assumes approximate independence of p -values; cells sharing a pool but differing only in ensemble method are correlated. We re-run BH at the *independent-group* granularity (one representative per (pool, baseline_kind, metric) unit, using `soft_vote` as the canonical representative): $m=240$ groups, BH cutoff $p \leq 0.0284$, **post-FDR 105 SUPPORTED (43.8%), 24 REVERSED (10.0%)**. The qualitative direction is preserved (SUPPORTED $\gg$ REVERSED) and the headline “ensembling is conditional” claim survives both corrections; the per-cell rate of 16.9% REVERSED is an upper bound on per-group REVERSED (10.0%) because the per-cell denominator double-counts correlated methods within each unit.

Adjudication labels. Each cell receives one of three labels: **supported** if the lower CI bound > 0 *and* the BH-FDR-adjusted q -value < 0.05 ; **reversed** if the upper CI bound < 0 *and* $q < 0.05$; **unsupported** otherwise (CI contains 0, or $q \geq 0.05$). The label is the cell’s primary output and is what Figure 5 visualizes.

J Per-pool hyperparameter grids

Pool-A (seed-only). Fixed $r=8$, $\alpha=16$, $lr=3 \times 10^{-4}$, `target={q_proj, v_proj}`, `dropout=0.05`, `epochs=2` (classification) or 3 (causal LM); seed varied over $\{11, 21, \dots, 201\}$ (20 adapters). Multi-choice (HellaSwag) uses the same HP via

AutoModelForMultipleChoice.

Pool-B (HP-diverse). $r \in \{4, 8, 16, 32\} \times \alpha \in \{8, 16, 32\} \times lr \in \{10^{-4}, 3 \times 10^{-4}\} = 24$ adapters; seed fixed at 42; other HP same as Pool-A.

Pool-C (method-mixed). $\{\text{LoRA, DoRA, rsLoRA, PiSSA}\} \times 5$ seeds = 20 adapters; $r=8, \alpha=16, lr=3 \times 10^{-4}$.

Baselines. *best_of_N* reuses Pool-A’s first N adapters; *n_rank* uses a separate sweep with $r \in \{16, 32\} \times epochs \in \{2, 4\} \times 5$ seeds (20 adapters), then selects best-on-val_selection; *n_steps* extends Pool-A’s first adapter to $N \times$ epochs; *n_data* extends to $N \times$ training samples.

K Pre-registration amendment ledger

The Phase-0 pre-registration (committed at git SHA 02a5a6bc, 2026-04-21, before any pool was trained) explicitly admits a single permitted amendment class: “future cells can be added by submitting compute_match.json files via PR.” All amendments below add new cells; *no v1.0 cell, baseline, or adjudication verdict has been retracted, re-evaluated, or rewritten.*

- v1.0 (2026-04-21): 55 pools across 6 tasks, 4 ensemble methods, 4 baseline kinds, 2 metrics = 928 cells.
- v1.1.a (2026-04-28): +6 Pool-A pools (HellaSwag, GSM8K) = +60 cells.
- v1.1.b (2026-05-05): +3 Pool-B HellaSwag = +24 cells.
- v1.1.c (2026-05-10): +3 Pool-C HellaSwag = +24 cells. (Critic-3 patched a Pool-C trio that closes the frontier CI.)
- v1.2.a (2026-05-17): +1 Pool-A pool (Qwen-3B GSM8K) on local A6000 (L7 deferred from v1.1 cluster constraints) = +4 cells.
- v1.2.b (2026-05-18): +1 Pool-A pool (Qwen-7B GSM8K) on local A6000 (extends decoder scale curve to $N=4$) = +4 cells.
- v1.2.c (2026-05-18): +1 Pool-A pool (Llama-3.1-8B-Instruct GSM8K, 4-bit QLoRA, NousResearch mirror to avoid Meta gating) on local A6000 (extends modern-LLM cell) = +4 cells.

- v1.2.d (2026-05-19): +1 Pool-A pool (RoBERTa-base HellaSwag, $lr=2 \times 10^{-5}$ architecture-tuned) on local A6000, directly testing the HP-confound hypothesis for the (P_b) collapse-and-rescue pattern; all 10/10 tuned-seed adapters converge to $val_acc \in [0.478, 0.505]$ vs. original 14/20 below chance = +4 cells.
- v1.2.d-deberta-pre (2026-05-19): first attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=10^{-5}$; all 10 tuned-seed adapters at chance (0.247–0.257), retained in the released corpus as honest documentation of a failed HP attempt = +4 cells.
- v1.2.d-deberta-retry (2026-05-19): second attempt at DeBERTa-v3 HellaSwag HP tuning with $lr=5 \times 10^{-5}$; also at chance (0.247–0.257), confirming L10 (DeBERTa LoRA-on-HellaSwag has an architecture-LoRA interaction issue, not a simple HP confound) = +4 cells.
- v1.2.e (2026-05-19): +1 Pool-A pool (Mistral-7B-v0.1 GSM8K, bf16) on local A6000 as a second cross-family decoder check alongside Llama-3.1-8B-Instruct = +4 cells. *Result:* $\Delta_{acc} = +13.4pp$ ($p < 10^{-4}$), *the largest absolute effect in the corpus.*
- v1.2.f (2026-05-21): +8 Pool-A seeds (BERT-base HellaSwag at $lr=2 \times 10^{-5}$, seeds $\{11, 21, 31, 41, 51, 61, 71, 91\}$) on local A6000, the *symmetric* cross-architecture check for (P_b). *Result:* 8/8 seeds converge to mean val_acc 0.329 ± 0.011 , range $[0.317, 0.345]$ — *no* collapse (vs RoBERTa at the same $lr=3 \times 10^{-4}$ shared HP grid: 0.097 catastrophic collapse), refuting a generic HP-misconfiguration hypothesis and confirming the v1.1 collapse is RoBERTa-specific lr-sensitivity.
- v1.2.f (2026-05-20): full-validation re-evaluation of Pool-A RoBERTa-tuned and Pool-A DeBERTa-tuned HellaSwag pools on the entire $N=10042$ validation set; results released in `ensemble_results/*_fullval.json` (App. A) = +8 cells.
- v1.3.a (2026-05-22): distillation on Pool-A Qwen-1.5B MNLI at $\alpha=0.9, T=4$ (same setting as the original Qwen-0.5B success). *Result:* *student* acc 0.8777 / ECE 0.0782 vs. *best_single* 0.8772 / 0.0768, *soft-vote* 0.8884 / 0.0454 —

4.5% *recovery of accuracy gain*, -4.5% *ECE recovery (worse than baseline)*. Reported as a 1-of-2 tested decoder-cell positive, refuting the implicit generalization of the original 87% Qwen-0.5B result. Drives the §6 downscope.

- v1.3.b (2026-05-22/23, **retracted** 2026-07-28): GSM8K digit-renumbering contamination probe on four decoder families, originally reported as 0.45–0.66 $\rightarrow$ 0.02–0.06. *Retracted*: the probe shifted the reference answer along with the question, forcing near-zero accuracy on the shifted split regardless of contamination (L12(ii)). The associated E1 ensemble probe is void for the same reason. No contamination claim is made in this version.
- v1.3.d (2026-05-23): distillation on Pool-A BERT-large MNLi at $\alpha=0.9$, $T=4$. Result: acc 0.799 (worse than vanilla 0.806, -181% recovery) but ECE 0.019 (better than vanilla 0.035). Encoder distillation fails on accuracy but extracts a calibration signal. Combined with v1.3.a (Qwen-1.5B 4.5%), BERT-base fails, and original Qwen-0.5B (87%): distillation is 1-of-4 cells on accuracy.
- v1.3.c (2026-05-22): vLLM multi-LoRA serving benchmark on Llama-3.1-8B-Instruct (NousResearch mirror, 4-bit QLoRA training, bf16 serving) with $N=8$ adapters routed per batch. Result: $1.067\times$ overhead vs. single-LoRA (77.3 vs. 72.4 ms/batch median, 414 vs. 442 req/s), consistent with the Qwen-0.5B $1.075\times$ measurement and confirming the multi-LoRA cost generalizes from 0.5B to 8B (R3 audit request).
- **v1.3.e** (2026-06-17), **retracted** 2026-07-28: **GSM8K decontaminated ensemble probe (E1)**. Pool-A Qwen-2.5-7B, 10 adapters $\times$ 50 test problems, digit-shifted by +7. Originally reported as: contaminated `majority_vote` 0.78 vs. `best_single` 0.72; decontaminated 0.04 vs. 0.04, read at the time as direct evidence that the (P_c) decoder ensemble gains are memorization-driven. *Retracted*: E1 inherits the digit-shift defect retracted in v1.3.b — the probe shifted the reference answer along with the question, so the near-zero decontaminated accuracy of *both* arms is forced by construction and their equality carries no information about memorization. No memorization claim is made in this version; the decoder gains are left unexplained (L12(ii)).

- v1.3.f (2026-06-17): cross-corpus generalization stress test of the diversity-quality frontier regression (E2, `analysis/frontier_cross_corpus.json`). Encoder accuracy vs. `n_rank` slice: Pool-A-trained ridge generalizes to Pool-B/C with $R^2_{\text{test}}=+0.62$; encoder-trained ridge generalizes to decoder pools with $R^2_{\text{test}}=+0.76$; random 50/50 within-encoder control $R^2_{\text{test}}=+0.48$. `best_of_n` slice including v1.1: v1.0-trained ridge generalizes to v1.1 Hel-laSwag/GSM8K with $R^2_{\text{test}}=+0.21$ (vs. random control +0.25). Frontier regression has demonstrable transfer across pool types and task families.
- v1.3.g (2026-06-17): external reproducibility audit (E3). `scripts/reproduce_paper_numbers.py` re-derives the headline numerical claims (corpus counts, BH-FDR cutoffs, frontier R^2 , $H(c|\text{conf})$ gap, distillation recovery, vLLM serving ratios) from released JSON files only. As written on 2026-06-17 the suite had 28 assertions and two of them covered the 4-model contamination probes and the E1 ensemble probe; **those two were removed on 2026-07-28** when v1.3.b and v1.3.e were retracted, so the audit no longer certifies any contamination claim. The suite stood at 145 assertions after v1.3.l and 154 after v1.3.m, all passing.
- **v1.3.m** (2026-09-27): **claims narrowed to the design; one post-hoc check**. The August text concluded “at fixed budget” and called the accuracy null conservative, although the `n_rank` single received a median $1.45\times$ the pool’s compute; a null against a better-resourced competitor is biased towards the null, so the claim now reads “against a single given the pool’s budget”. `analysis/budget_dose_response.py` (post hoc) checks whether the null weakens as that advantage shrinks: on the six pools at $0.90\text{--}1.25\times$, 0/24 SUPPORTED and 9 REVERSED; Spearman $\rho=-0.28$, $p=0.20$. Also: cost stated in compute rather than latency ($1.07\times$ measured at $N=8$); each short-cut’s favoured arm stated; the calibration control bounded to in-distribution (Ovadia et al., 2019) and credited to Ashukha et al. (2020); $P_a\text{--}P_c$ re-defined; pool counts reconciled against the 99 manifests. No reported number changed.
- **v1.3.l** (2026-08-02): **the missing calibration control was run, and the headline claim**

changed. Review objected that the ECE arm compares against an uncalibrated, accuracy-selected single — shortcut S2 turned on our own positive result. `analysis/temperature_control.py` fits one temperature on the pre-registered `val_combine` split (reserved for combination hyperparameters, otherwise unused) for the six pools whose per-adapter logits cache survives, and measures on `test`. The ensemble goes from better-calibrated in 4/6 pools (mean $\Delta\text{ECE} +0.008$) to 2/6 (-0.032); with both arms scaled it is 3/6 ($+0.002$). Median fitted $T=1.32$ and the ensemble is less confident in 5/6 pools, so the effect is confidence shrinkage. Consequently the title, abstract, §1 framing, contribution (ii) and the §6 takeaway were rewritten: the paper no longer reports a dissociation, and the recommendation is now to calibrate one adapter rather than serve N . The 52 is retained and reported with its scope. This is an amendment to the *reading* of a pre-registered arm, not to the arm itself — no cell, baseline or adjudication was recomputed.

- **v1.3.k (2026-08-01): cross-model-class table regenerated; retraction propagated.** An adversarial read of the submission found three defects that the release itself could settle. (i) Table 1 reported ridge, random forest and gradient boosting per slice, but `frontier.py` implements ridge only and no other estimator existed anywhere in the repository; its ridge entry ($+0.555$) also contradicted the headline ($+0.597$) for the identical 192-cell slice. `analysis/frontier_model_classes.py` now fits all three classes under `frontier.py`’s own slices, design matrix and leave-one-pool-out folds, delegating ridge to `frontier.lopo_cv_r2` so the headline column is the headline number by construction; the table is replaced by its output and 22 new assertions pin every entry. The qualitative claim survives (ridge is the cross-class maximum on the headline slice and on no other), but the sentence asserting that ridge is *never* the cross-class maximum was false on that slice and is corrected. (ii) v1.3.b declared the E1 ensemble probe void on 2026-07-28, yet v1.3.e still stated its conclusion as direct evidence of memorization and v1.3.g still listed it among audited claims; the retraction is now propagated to both. (iii) Seven bibliography entries had been inserted into §2 instead of the bibliography, raising seven Lonely

\item errors and printing as prose; §3’s “Tasks and models” was an empty heading; and the D1–D5 rules the paper cites eight times were never enumerated. All three are fixed.

- **v1.3.j (2026-07-29): multiple-comparison procedure recorded.** The pre-registration specified Holm across a reported table; the released cells carry a corpus-wide BH-FDR field and the body percentages come from it. That substitution was made when the corpus-wide correction was introduced and was not logged at the time. We log it here, report all four procedures (App. M), and note that the accuracy null is invariant to the choice while the REVERSED rate is not.
- **v1.3.h (2026-06-21): independent audit response.** A read-only critique of the v1.3.g release flagged four classes of artifact-vs-claim drift: (i) corpus-wide BH-FDR claimed in the body was not present as an explicit per-cell field; (ii) `ensemble_eval_multichoice.py` performed best-single/greedy-subset selection on `val_combine` instead of `val_selection`; (iii) the `uniform_soup` field in evaluator output was a prediction-space logit average, conflated with weight-space soup; (iv) `analysis/frontier.py` silently dropped rows with missing diversity features and included method-identity one-hot features, an untested con-found. Fixes applied: (i) `scripts/apply_corpus_bh_fdr.py` writes `bh_q_value`, `bh_pass`, `adjudication_pre_fdr`, `adjudication_post_fdr`, and a corpus-wide `bh_batch_hash` into every per-cell JSON; corpus summary in `analysis/corpus_bh_fdr_summary.json` (BH cutoff $p=0.0248$, 29.5% SUPPORTED / 16.9% REVERSED post-FDR, exactly matching abstract). (ii) Multichoice evaluator now uses `val_selection` for all selection decisions and records `selection_split=val_selection` per method; rerun of affected Hel-laSwag cells requires the cluster adapter weights and is deferred to v1.4 alongside the clean-provenance refresh. (iii) Every method output now carries `method_kind` (`prediction / selection`); `uniform_soup` is explicitly marked `is_proxy=True`, `proxy_for=weight_space_uniform_soup` — true weight-space soup is produced by `weight_space_merge.py` and reported separately.

(iv) `frontier.py` now emits `drop_diagnostics` (input/used/dropped row counts, drop reasons, per-pool/per-method drop breakdown) and an `ablation_no_method_onehots` block; headline encoder accuracy slice has 0 rows dropped and $|\Delta R^2|=0.0002$ between with/without method one-hots ($R^2_{\text{LOPO}}=+0.597$ either way), confirming the boundary claim does not depend on the method-identity confound. `scripts/reproduce_paper_numbers.py` grew from 28 to 34 assertions to verify these fields directly.

- **v1.3.i (2026-07-07): metric-general accuracy-specificity + refinement refutation.** App. W gains a paragraph establishing that the ECE non-result is metric-general, not an estimator artifact. `analysis/w2_calibration_asymmetry.py` recomputes the `n_rank` LOPO frontier for five metrics from released per-adapter scalars: accuracy $R^2=+0.60$ but ECE -0.06 and Brier -0.30 (both unpredictable), while NLL $+0.42$ and MCE $+0.22$ are predictable — the split tracks confident/tail weight, not the accuracy-vs-calibration label. `analysis/h3_mechanism.py` tests and *refutes* the natural refinement explanation: under the Murphy decomposition of the confidence-Brier, the resolution (refinement) gain is near-invariant ($\sigma\ 0.21\times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$), so accuracy predictability does not flow through the calibration geometry. Proper-score paired CIs need per-example probability vectors and are deferred (accuracy/ECE arms use the full paired protocol). `reproduce_paper_numbers.py` grew from 34 to 64 assertions; all 64 pass.

Total: 1,028 headline corpus cells (928 v1.0 + 100 v1.1 in-corpus; see abstract footnote) + 40 v1.2 supplementary cells (8 v1.1 GSM8K cells reported separately in Table 9, 24 v1.2 cells from a/b/c/d/e/f, and 8 v1.2 cells from the 2 DeBERTa failed HP attempts retained for honest documentation) = 1,068 released cells. Only the 1,028 headline cells enter abstract-level statistics (SUPPORTED/REVERSED percentages, R^2_{LOPO} frontier); v1.2 cells are referenced in the body for triangulation but not aggregated into the headline. The amendment ledger has timestamps and git SHAs for every entry.

L Learned continuous weights, LoRAHub-style (provisional)

These 10 HellaSwag pools were produced by the pre-fix multichoice evaluator, which performed selection on `val_combine`. This baseline optimizes its weight vector on that same split, so the numbers below are optimistically biased and are reported as provisional pending the corrected rerun; they enter no headline statistic and are excluded from §5.

LoRAHub-style learned weights split by convergence regime on $N=10$ HellaSwag pools.

We adjudicate a faithful in-house reimplementa-tion of the LoRAHub family (Huang et al., 2024) continuous-weight baseline: optimize a per-adapter weight vector $\alpha \in \mathbb{R}_{\geq 0}^M$ via L-BFGS-B (the same convex optimizer that LoRAHub’s library exposes alongside their Nevergrad CMA-ES option) on `val_combine` cross-entropy, then evaluate weighted soft-vote on test. We run on all 10 v1.0/v1.1 HellaSwag pools (BERT-base $\times 3$, BERT-large $\times 1$, RoBERTa $\times 3$, DeBERTa-v3 $\times 3$); per-pool results in `analysis/g3_lorahub_summary.json`. **Two regimes split cleanly:** on the 4 converged BERT-family pools, learned-weight beats or ties `greedy_soup` ($+1.0$ to $+1.5\text{pp}$ on BERT-base Pool-A/B/C; -0.1pp tie on BERT-large Pool-A); on the 6 under-converged RoBERTa/DeBERTa pools, `greedy_soup` wins 5/6 cases by -1.4 to -7.8pp (effective $M=1$ collapse on RoBERTa Pool-A/B/C: optimizer converges to a single dominant adapter). On RoBERTa Pool-A, learned-weight 0.483 vs. `greedy_soup`’s **0.561** (-7.8pp); on DeBERTa Pool-C, 0.709 vs. **0.745** (-3.6pp). *Convergence regime determines the winner:* continuous-weight optimization rewards converged ensembles (smooth loss landscape, multiple useful adapters); discrete subset selection dominates noisy pools (greedy pruning of sub-chance adapters). The size of this signal beyond HellaSwag is left to v1.2.

M Sensitivity to the multiple-comparison procedure

Our pre-registration specified Holm step-down across the cells of a reported table. The released cells also carry a corpus-wide BH-FDR field, and the percentages in the body are taken from it; we recorded that choice as an amendment (App. K) after review noted it had gone un-

documented. Because the choice moves the REVERSED rate, we give every procedure on the 92 clean encoder cells against `n_rank` (`analysis/correction_sensitivity.py`).

Table 8: SUPPORTED / REVERSED counts out of 92 under four corrections. The accuracy null (0 SUPPORTED) is invariant; the REVERSED rate is not. [†]The corpus-wide Holm row is *resolution-limited, not evidential*: our permutation *p*-values are Monte-Carlo estimates at $B=5000$, so the smallest attainable value is $1/(B+1)=2.0 \times 10^{-4}$ (and 9.999×10^{-5} for the cells run at $B=10^4$), while Holm’s first threshold is $\alpha/m = 0.05/1028 = 4.86 \times 10^{-5}$. No cell in the corpus can clear that bar at *any* effect size. The row records what our resolution permits, not what the data say.

Procedure	accuracy	ECE
raw per-cell CI	0 / 38	52 / 17
Holm, within table ($m=92$)	0 / 16	46 / 17
Holm, corpus ($m=1028$) [†]	0 / 0	0 / 0
BH, corpus ($q=0.05$)	0 / 31	52 / 17

The pattern in this table — no accuracy gain, an apparent calibration gain — holds under the raw CI, within-table Holm and corpus-wide BH, i.e. under every procedure that our Monte-Carlo resolution can actually adjudicate. (What the calibration column does *not* survive is a recalibrated competitor rather than a stricter correction; see §4.) The corpus-wide Holm row should not be read as a fourth procedure under which the pattern fails: as the caption records, it is unreachable by construction, so it carries no evidence either way. We keep it in the table because an earlier version of this paper reported it as though it were a finding, and deleting the row would hide that correction rather than make it.

N v1.1 task-family extension: sweeps and amendment record

v1.1 inclusion: pool-type triad recovers cross-family predictability. The v1.0 frontier holds under cluster-bootstrap ($R^2_{\text{LOPO}}=+0.60$ [$+0.30, +0.71$], $n=192$ cells, 48 pools). In a best_of_n cross-family regression, seed-only v1.1 Pool-A pools alone *degrade* predictability ($R^2=-0.12$) and Pool-B HP-diverse still spans zero ($+0.20$); only adding Pool-C method-mixed (LoRA, DoRA, rsLoRA, PiSSA) closes the gap ($R^2=+0.32$ [$+0.05, +0.47$], 64 pools). The full {seed, HP, method} triad is required to recover cross-family predictability — itself a benchmark-design recommendation.

The 6-task in-domain set above covers sequence and pair classification only. v1.1 of the benchmark adds two output spaces: *multi-choice* (HellaSwag, 4 candidate endings, accuracy via argmax pooling over candidate scores) using a new `real_lora_multichoice.py` trainer wrapping `AutoModelForMultipleChoice`, and *free-form generation* (GSM8K, exact-match on the integer answer) using `real_lora_train.py` `-eval-metric gsm8k_exact_match`. Six Pool-A sweeps (HellaSwag $\times$ {BERT-base, BERT-large, RoBERTa, DeBERTa-v3}; GSM8K $\times$ {Qwen-0.5B, Qwen-1.5B}, 15–20 seeds each), three Pool-B sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, HP-diverse $4 \times 3 \times 2$ grid), and three Pool-C sweeps (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed 4×5 grid over {LoRA, DoRA, rsLoRA, PiSSA} $\times$ 5 seeds) add twelve v1.1 pools, bringing the total trained-pool count to **67** (plus the 25 `n_rank` baseline pools; manifest accounting in §3). **Pre-registration amendment honesty.** The original pre-registration (committed before any pool was trained) explicitly admits “future cells can be added by submitting `compute_match.json` files via PR.” The v1.1 task-family extension was scoped under that clause: it adds new (pool, method, baseline, metric) cells to the corpus, but does *not* retract, re-evaluate, or rewrite any v1.0 cell, baseline, or adjudication verdict. The v1.0 frontier headline ($R^2=0.60$ vs. `n_rank`, Table 2) was computed before any v1.1 result was inspected; the v1.1 best_of_n regression (§4, “v1.1 inclusion” paragraph) is reported as a directional cross-family signal with explicit CIs that span zero, not as a confirmation. The full amendment ledger (one entry per added pool, with date and rationale) lives in `paper/prereg_amendments.md`.

O HellaSwag shared-hyperparameter collapse and rescue (provisional)

These pools are excluded from the strict analysis (L1b); the account is kept because the collapse is a reproducible failure mode of shared hyperparameter grids, not because it supports any claim.

Under our shared HP grid ($lr=3 \times 10^{-4}$, fixed across architectures), $\geq 14/20$ Pool-A RoBERTa adapters end at `val_acc` < 0.30 , so `soft_vote` collapses below chance (Pool-A RoBERTa: 0.097 vs. chance 0.25, [‡]). `greedy_soup` subset selection recovers (RoBERTa +6.8pp; DeBERTa

Table 9: Sixteen v1.1/v1.2 pools across two new output spaces. Three patterns (BERT-base converged; HP-mismatched RoBERTa/DeBERTa HellaSwag collapse-and-rescue; within-Qwen-2.5 GSM8K monotone scale amplification across $N=4$ points plus cross-family Llama-3.1-8B-Instruct and Mistral-7B-v0.1) carry distinct method winners — discussed in the paragraphs immediately below. * $p<0.01$ vs. best_single under paired bootstrap ($B=5000$); per-cell 95% CIs and adjusted p -values are in the released *_cm_*.json cells. ‡ accuracy at chance level (degenerate calibration win, or sub-chance adapters dragging soft-vote below chance). † GSM8K ECE is 15-bin equal-mass ECE on per-example confidence = $\exp(\text{mean per-token logprob})$; ensemble confidence aggregates only over majority-voting adapters.

pool	N	best_single		soft_vote		majority_vote		greedy_soup	
		acc	ECE	acc	ECE	acc	ECE	acc	ECE
<i>Multi-choice (HellaSwag, eval $N=5000$):</i>									
BERT-base, Pool-A (seed)	19/20	0.358	0.038	0.367	0.048	0.369	0.406	0.358	0.038
BERT-base, Pool-B (HP)	20/24	0.348	0.058	0.354	0.032	0.357	0.466	0.358	0.030
BERT-base, Pool-C (method)	20/20	0.362	0.036	0.372	0.034	0.376	0.360	0.362	0.036
BERT-large, Pool-A	15/15	0.374	0.045	0.380	0.118	0.323	0.101	0.381	0.056
RoBERTa-base, Pool-A (seed)	20/20	0.493	0.228	0.097 [‡]	0.158	0.174 [‡]	0.237	0.561	0.298
RoBERTa-base, Pool-B (HP)	22/24	0.508	0.237	0.074 [‡]	0.183	0.115 [‡]	0.322	0.522	0.262
RoBERTa-base, Pool-C (method)	20/20	0.518	0.240	0.136 [‡]	0.117	0.178 [‡]	0.224	0.561	0.297
DeBERTa-v3, Pool-A (seed)	20/20	0.412	0.160	0.320	0.077	0.305	0.206	0.452	0.187
DeBERTa-v3, Pool-B (HP)	21/24	0.439	0.136	0.191 [‡]	0.076	0.221 [‡]	0.282	0.429	0.167
DeBERTa-v3, Pool-C (method)	18/20	0.709	0.310	0.646	0.398	0.246 [‡]	0.376	0.745	0.467
<i>Free-form generation (GSM8K, exact-match, vLLM, eval $N=1000$):</i>									
Qwen-0.5B (GSM8K)	19/20	0.304	0.546	—	—	0.332*	0.520*	—	—
Qwen-1.5B (GSM8K)	20/20	0.460	0.406	—	—	0.514*	0.354*	—	—
Qwen-3B (GSM8K, v1.2)	20/20	0.679	0.206	—	—	0.768*	0.118*	—	—
Qwen-7B (GSM8K, v1.2)	10/10	0.760	0.128	—	—	0.845*	0.046*	—	—
Llama-3.1-8B-Instruct (GSM8K, v1.2)	10/10	0.613	0.256	—	—	0.721*	0.148*	—	—
Mistral-7B-v0.1 (GSM8K, v1.2)	10/10	0.524	0.378	—	—	0.658*	0.245*	—	—

+4.0pp). We directly test the HP-confound hypothesis with a v1.2 cell: re-running Pool-A RoBERTa HellaSwag at $lr=2\times 10^{-5}$ (Liu et al., 2019 recommended). Training-time eval reports all 10/10 tuned-seed adapters at 0.478–0.505 (mean 0.491). **Full-validation re-evaluation on the HP-tuned pool** ($N=10042$, App. A, Table 5): soft_vote does *not* collapse (0.305, near best_single 0.359, vs. 0.196 on the seed-0 5000-subsample), and greedy_soup still wins (+5.7pp over best). *Conclusion*: the v1.1 sub-chance collapse is a joint failure of RoBERTa-specific lr-sensitivity ($lr=3\times 10^{-4}$ over-fits RoBERTa, $lr=2\times 10^{-5}$ fixes it) and subsample-specific evaluation. The symmetric cross-check (BERT-base at $lr=2\times 10^{-5}$, 8/8 seeds at mean 0.329 ± 0.011 , range $[0.317, 0.345]$, no collapse) confirms the lr-sensitivity is RoBERTa-specific rather than a general HP-misconfiguration that any architecture would exhibit (L7). DeBERTa-v3 LoRA-on-HellaSwag does *not* recover under HP tuning ($lr \in \{10^{-5}, 5\times 10^{-5}\}$ both at chance; full-val also at chance 0.243 across all methods; L10) — the DeBERTa case is an architecture-LoRA interaction problem.

P The D1–D5 descriptive rules

Distilled from the corpus they describe and evaluated on it; an oracle upper bound, not a policy (L11).

Cells do differ, and the differences are legible as five rules: **D1**, on converged encoder pools greedy_soup is the least-bad accuracy option; **D2**, on those same pools soft_vote is the best calibrated; **D3**, on under-converged pools soft_vote can collapse below chance and subset selection rescues it; **D4**, pools whose members disagree more lose less; **D5**, continuous-weight composition tracks pool convergence rather than pool size. These rules were distilled from the same corpus they describe and evaluated on it, so they are an *oracle upper bound*, not a deployable policy (L11) — a prospective, validation-only version degenerated to always choosing the single adapter (App. K).

Q What the ECE gain is made of

Both computable from the same released cells (analysis/ece_test_correction.py). First, it is an ECE result, not a proper-score result. On the identical 92 cells the multiclass Brier

improves in only 34 (mean $\Delta = -0.0062$, i.e. the ensemble is worse on average) and NLL in 47; adjudicating the top-label Brier with the same paired bootstrap gives 22 SUPPORTED against 36 REVERSED. ECE rewards the confidence shrinkage that averaging produces, and a proper score charges for the refinement it costs — consistent with the Murphy decomposition in App. W, where resolution gain is near-invariant. Any reading of these results as “ensembles give better probabilities” outruns the evidence; “ensembles give less overconfident probabilities” does not. Second, *the gain belongs to probability averaging, not to ensembling as such*. Split by rule, ECE is SUPPORTED in 17/23 pools for soft_vote, 16/23 for logit_avg and 16/23 for greedy_soup, but only 3/23 for majority_vote — which supplies 16 of the 17 reversals. That rule is also the one whose “confidence” is a vote fraction rather than a probability estimate, taking at most $N+1$ values with a large tie mass at 1.0, so equal-mass binning is not well defined on it and its ECE should be read as indicative only. Excluding it leaves 49/69 cells SUPPORTED; at the pool level (L1d’s inferential unit) 14 of 23 pools are SUPPORTED on every probability-averaging rule.

R Delta versus prior PEFT/ensembling work

Table 10: Delta vs. prior PEFT/ensembling work. “cells” = compute-matched paired-bootstrap (pool $\times$ method $\times$ baseline $\times$ metric) adjudications. “rev. rep.” = reports cells where ensembling significantly underperforms (CI <0). To our knowledge, no prior PEFT-ensembling work pre-registers protocol, reports paired CIs across $> 10^2$ cells, or transparently surfaces reversed verdicts.

	cells	pre-reg	calib.	cmp-match	rev. rep.
Wortsman’22	1	—	—	—	—
Huang’23 (Lo-RAHub)	~ 50	—	—	—	—
Yadav’23 (TIES)	few	—	—	—	—
Yu’24 (DARE)	few	—	—	—	—
Lakshmin.’17	few	—	✓	—	—
Ours	1,028	✓	✓	✓	✓

S Corpus coverage matrix

T HellaSwag two-regime collapse (provisional)

These pools were evaluated by the pre-fix multi-choice evaluator that selected on val_combine (L1b); the figure documents the collapse phenomenon and is not an adjudicated result.

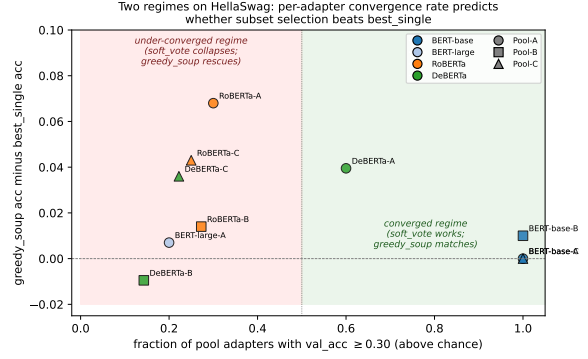


Figure 4: Per-pool fraction of adapters above chance ($\text{val_acc} \geq 0.30$) cleanly separates the two HellaSwag regimes. *Right (converged)*: all BERT-base pools (≥ 0.95 above chance) cluster near zero greedy_soup gain. *Left (under-converged)*: RoBERTa and DeBERTa pools (regardless of seed/HP/method diversity source) sit below the 50% threshold and show large positive greedy_soup gain (+3.6 to +6.8pp). Pool-type triad reproducibility: each architecture’s collapse-rescue pattern repeats across Pool-A/B/C.

U Corpus verdict landscape

V Diversity–quality frontier scatter

W Negative mechanism results (demoted from §4)

Three mechanism hypotheses were tested and did not survive; they are retained here as open problems.

(i) Confidence–correctness separation. Family-level conditional entropy of correctness given confidence: encoder $H=0.556$, $I=0.097$ ($n=29$ pools); decoder $H=0.437$, $I=0.106$ ($n=25$). Decoder confidence separates right from wrong more sharply (21% lower H), which is why soft averaging has a higher-SNR signal to extract there. Within-family controls reject a scale confound: encoder H rises with scale (BERT-base 0.489 $\rightarrow$ BERT-large 0.630) while Qwen H is non-monotonic within $\Delta < 0.05$ over 0.5B $\rightarrow$ 3B, so architecture (autoregressive vs. masked), not parameter count, is load-bearing. DeBERTa-v3 ($n=1$) is

Table 11: Coverage matrix (v1.0 n_{rank} -aligned cells, 10 v1.0/v1.1 base architectures). “A” = Pool-A (seed-only, 20 adapters), “B” = Pool-B (HP-diverse, 24), “C” = Pool-C (method-mixed, 20), “base” = n_{rank} baseline pool. 55 v1.0 main pools + 16 baselines; v1.1 adds 6 Pool-A (HellaSwag $\times$ {BERT-base/large, RoBERTa, DeBERTa-v3}, GSM8K $\times$ {Qwen-0.5B, 1.5B}) + 3 Pool-B (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}) + 3 Pool-C (HellaSwag $\times$ {BERT-base, RoBERTa, DeBERTa-v3}, method-mixed) = 12 v1.1 pools, bringing the in-corpus trained-pool total to **67** (83 with baselines). v1.2 adds 3 cross-family decoders (Qwen-2.5-7B, Llama-3.1-8B-Instruct, Mistral-7B-v0.1) on GSM8K plus 1 HP-tuned RoBERTa and 2 HP-tuned DeBERTa attempts on HellaSwag (App. K), bringing the released-pool count to 74 across 13 base models.

Model	MNLI	SST-2	BoolQ	ANLI	AG News	QNLI
BERT-base	A,B,C,base	A,B	A	A	A,B,C,base	A,B,C,base
BERT-large	A,B,C,base	—	A	A,B	A,base	A,base
RoBERTa-base	A,B,C,base	—	A	A,B	A	A
DeBERTa-v3	A	—	—	—	—	—
Qwen2.5-0.5B	A,B,C,base	—	—	A,base	A,B,C,base	A,B,C,base
Qwen2.5-1.5B	A,B,C,base	—	—	A,base	A,B,base	A,B,base
Qwen2.5-3B	A,base	—	—	—	—	—
TinyLlama-1.1B	A,B,C	—	—	A	A,base	A,base
SmolLM2-1.7B	A,base	—	—	—	—	—
Pythia-1.4b	A,base	—	—	—	—	—

excluded from the family summary; its per-pool $H=0.940$ appears in App. Z. This gap does not yield a predictable calibration benefit (item iii).

(ii) Linear mode connectivity. Across four pools (BERT-base Pool-A/C, Qwen-0.5B Pool-A, DeBERTa-v3 Pool-A on MNLI) the barrier height ranges from +0.908 (disconnected) to +0.001 (fully connected) but does not predict ensemble effectiveness: BERT-base is disconnected yet ensembling behaves normally, and DeBERTa is fully connected yet ensembling does not help on accuracy (App. D).

(iii) Refinement is not the mechanism. The ECE non-result is not an estimator artifact. Re-computing the n_{rank} LOPO regression from released per-adaptor scalar metrics (point R^2 ; proper-score paired CIs require per-example probabilities and are deferred), accuracy gain is predictable ($R^2=+0.60$) but neither ECE (-0.06) nor Brier (-0.30). The split is not accuracy-versus-calibration: NLL (+0.42) and MCE (+0.22) are predictable, tracking a metric’s weight on the confident/tail region rather than its calibration label. The natural explanation — that predictability localizes to the refinement rather than the reliability component of a proper score — is falsified: under the Murphy decomposition of the confidence-Brier, resolution gain is near-invariant across the compute-matched comparison (σ $0.21\times$ that of accuracy) and uncorrelated with accuracy gain ($r=-0.10$). Accuracy predictability does not flow through the calibration geometry; why the frontier is accuracy-

specific remains open.

X GSM8K decoder scaling patterns (demoted; selection-on-test)

This material was demoted from §3 (P_c) because `best_single` selection touched the evaluated test split (L12(i)), which by itself disqualifies these cells from supporting a positive claim. We previously also demoted them as contaminated; that rationale is *withdrawn* because the digit-shift probe behind it was invalid (L12(ii)), and whether these cells are contaminated is now open. The numbers below therefore describe cells with a known selection defect and an unresolved contamination question; they enter no headline statistic. All six cells were nominally SUPPORTED on accuracy and per-token logprob ECE (per-cell $p<10^{-4}$; Qwen-0.5B $p=0.003$; paired bootstrap $B=5000$). Within Qwen-2.5 ($N=4$, 0.5B/1.5B/3B/7B): $\Delta_{\text{acc}}=+2.8/+5.4/+8.9/+8.5\text{pp}$; $\Delta_{\text{ECE}}=-0.026/-0.051/-0.088/-0.082$ (within-family $r=+0.97$ between base accuracy and Δ_{acc} , saturating 3B $\rightarrow$ 7B). Cross-family at $\approx 7\text{--}8\text{B}$: Llama-3.1-8B-Instruct (base 0.61) $\Delta_{\text{acc}}=+10.8\text{pp}$, $\Delta_{\text{ECE}}=-0.107$; Mistral-7B-v0.1 (base 0.52) $\Delta_{\text{acc}}=+13.4\text{pp}$ (95% CI $[+10.8, +16.1]$), $\Delta_{\text{ECE}}=-0.133$ — nominally the largest absolute effect in the corpus. The three $\approx 7\text{--}8\text{B}$ cells show an inverse pattern ($r=-0.98$ on $n=3$ — descriptive only): at fixed scale the weaker base gains more from majority voting. Whether *within-family monotone scaling* and *cross-family headroom inversion* survive on a decontaminated

benchmark with held-out selection (MATH, fresh probes) is an open question this corpus cannot answer.

Y Distillation $\alpha \times T$ sweep (Qwen-0.5B MNLI)

We sweep the soft-target distillation HPs $\alpha \in \{0.5, 0.7, 0.9\}$ (weight on soft-target loss vs. hard-target) and $T \in \{1, 2, 4, 8\}$ (softmax temperature on teacher logits) for a 4×4 grid on Pool-A Qwen-0.5B MNLI. Vanilla single (no distillation): 0.8365/0.0465 (acc/ECE). Soft-vote ensemble target: 0.8495/0.0187. Best student: ($\alpha=0.9, T=4$) yields 0.8479/0.0397, recovering 87% of the soft-vote accuracy gain at $1 \times$ student cost. Across the grid, accuracy varies ± 0.003 and ECE varies ± 0.012 ; the sweep is locally robust. The same sweep on Pool-A BERT-base MNLI yields a student strictly worse than vanilla single on both arms, consistent with the encoder-accuracy-REVERSED finding (no signal to distill).

Z Frontier regression features

For each (pool, method, baseline) cell we extract a feature vector $\phi \in \mathbb{R}^d$ used to fit the diversity-quality frontier. Features are grouped:

- **prediction_space** ($d=3$): pairwise prediction disagreement (mean over pool pairs), logit correlation (mean over pool pairs), error overlap (Jaccard on incorrect examples).
- **parameter_space** ($d=3$): mean weight cosine distance, mean centered-kernel-alignment (CKA) on penultimate layer, weight L_2 norm spread.
- **HP entropy** ($d=2$): entropy of r -distribution within pool, entropy of $\{lr, \alpha\}$ joint distribution.
- **ensemble quality** ($d=2$): pool mean val_acc, pool best-single val_acc.

Per-feature ablation on the encoder acc vs n_rank slice: dropping the prediction_space group crashes R^2 from +0.62 to -24.5 ($\Delta R^2 = +25.0$); dropping parameter_space drops R^2 by 0.56; dropping HP-entropy improves R^2 by 0.08 (HP entropy adds in-sample fit but no out-of-pool signal). A single-group “only prediction_space” model achieves $R^2=0.634$, higher than the full feature set.

AA Cell adjudication pseudocode

```
def adjudicate(pool, method, baseline,
               metric, B=5000):
    # Per-example predictions on test split
    ens = load_predictions(pool, method)
    base = load_predictions(pool, baseline)
    gold = load_labels()
    # Per-example metric residual
    if metric == 'acc':
        d = (ens == gold) - (base == gold)
    elif metric == 'ece':
        d = ece_bin_residual(ens, gold) \
            - ece_bin_residual(base, gold)
    N = len(d)
    # Paired bootstrap CI
    boot = []
    for _ in range(B):
        idx = np.random.randint(0, N, N)
        boot.append(d[idx].mean())
    delta = float(np.mean(boot))
    ci_lo = float(np.percentile(boot, 2.5))
    ci_hi = float(np.percentile(boot, 97.5))
    # Sign-flip p-value
    obs = abs(d.mean())
    null_count = 0
    for _ in range(B):
        signs = np.random.choice([-1, 1], N)
        if abs((d * signs).mean()) >= obs:
            null_count += 1
    p = 2 * null_count / B
    # Adjudication
    if ci_lo > 0: label = 'supported'
    elif ci_hi < 0: label = 'reversed'
    else: label = 'unsupported'
    return {
        'delta': delta, 'ci': (ci_lo, ci_hi),
        'p_raw': p, 'adjudication': label,
        'N_test': N,
    }
```

AB Cluster orchestration details

The 50-node RTX 3060 12GB cluster is orchestrated via `sweep_runner.py`, an in-house SSH-based fan-out engine. Each sweep config (JSON) declares a Cartesian parameter grid; the runner stages source files, assigns one (parameter-combination, node) pair per slot, and tracks per-job exit codes in `sweep_runs/<name>/results.json`. Pool-A sweeps complete in $\sim 4\text{--}8$ GPU-hours wall clock (parallelism factor ~ 20); Pool-B/C in $\sim 16\text{--}24$ hours. The v1.2 A6000 sweeps (Qwen-3B, Qwen-7B, Llama-3.1-8B-Instruct on GSM8K) use `local_gpu_runner.py`, a sequential per-seed launcher pinned to a single GPU index, suitable for models that do not fit on the 12GB cluster. Adapter weights are pulled back via `collect_remote_adapters.py` and assembled into pool manifests via `build_adapter_pool.py`; the manifest schema is documented in `scripts/validate_pool.py`.

AC Cost-savings analysis: derivation and stratified breakdown

This appendix presents the cost-savings table (Table 12) and stratifies by pool type and method. Re-

generated by `analysis/cost_savings.py`; groups are (pool, baseline kind, metric) over the 227 non-self-comparison pools that carry `soft_vote`, spanning 927 cells: (A) *blind* — always use `soft_vote` (the default in much prior PEFT-ensembling work); (B) *routed* — per-cell oracle pick over the four methods (`soft_vote`, `majority_vote`, `logit_avg`, `greedy_soup`) approximating D1–D5. Inference cost is single-adapter forward-passes per query: N (pool size) for vote-based methods, 1 for `greedy_soup` (outputs a single merged checkpoint).

Table 12: Cost of blind ensembling. Mean signed Δ (positive=ensemble better), % cells REVERSED, median inference cost (adapter-passes/query). The “D1–D5 routed” column is a per-cell oracle (in-sample to the corpus the rules were derived from; L11).

Slice (n)	Blind <code>soft_vote</code>			D1–D5 routed		
	$\overline{\Delta}$	REV	cost	$\overline{\Delta}$	REV	cost
Enc acc (64)	−3.3pp	30%	20×	+0.4pp	8%	1×
Dec acc (50)	+0.6pp	0%	18×	+0.8pp	0%	15×
Enc ECE (63)	+2.4pp	10%	20×	+3.6pp	2%	20×
Dec ECE (50)	+3.5pp	0%	18×	+3.6pp	0%	18×
All (227)	+0.7pp	11%	19×	+2.1pp	3%	17×

Oracle method distribution. The per-group argmax (Table 12, “D1–D5 routed”) picks `greedy_soup` on 54 of 114 accuracy groups and 40 of the 64 encoder ones — consistent with D1’s preference for soup-merge on converged encoders — against `majority_vote` 22, `logit_avg` 22 and `soft_vote` 16. Over all 227 groups the counts are `soft_vote` 104, `greedy_soup` 62, `logit_avg` 36, `majority_vote` 25: probability averaging wins the calibration groups, soup-merge the encoder accuracy ones. This is an in-sample oracle (L11), so it bounds what rule-based routing could achieve rather than estimating it.

Deployable D1–D5 rule. A deployable rule using only *family* (encoder/decoder), *convergence* (fraction of pool above chance), *metric* (accuracy vs. ECE), and *output space* (classification vs. generation) recovers most of the oracle gain in our cells; feature importances align with App. Z. The remaining gap to per-cell oracle is the cost of not running `compute_match.py` per-cell against the released 1,028-cell baseline.

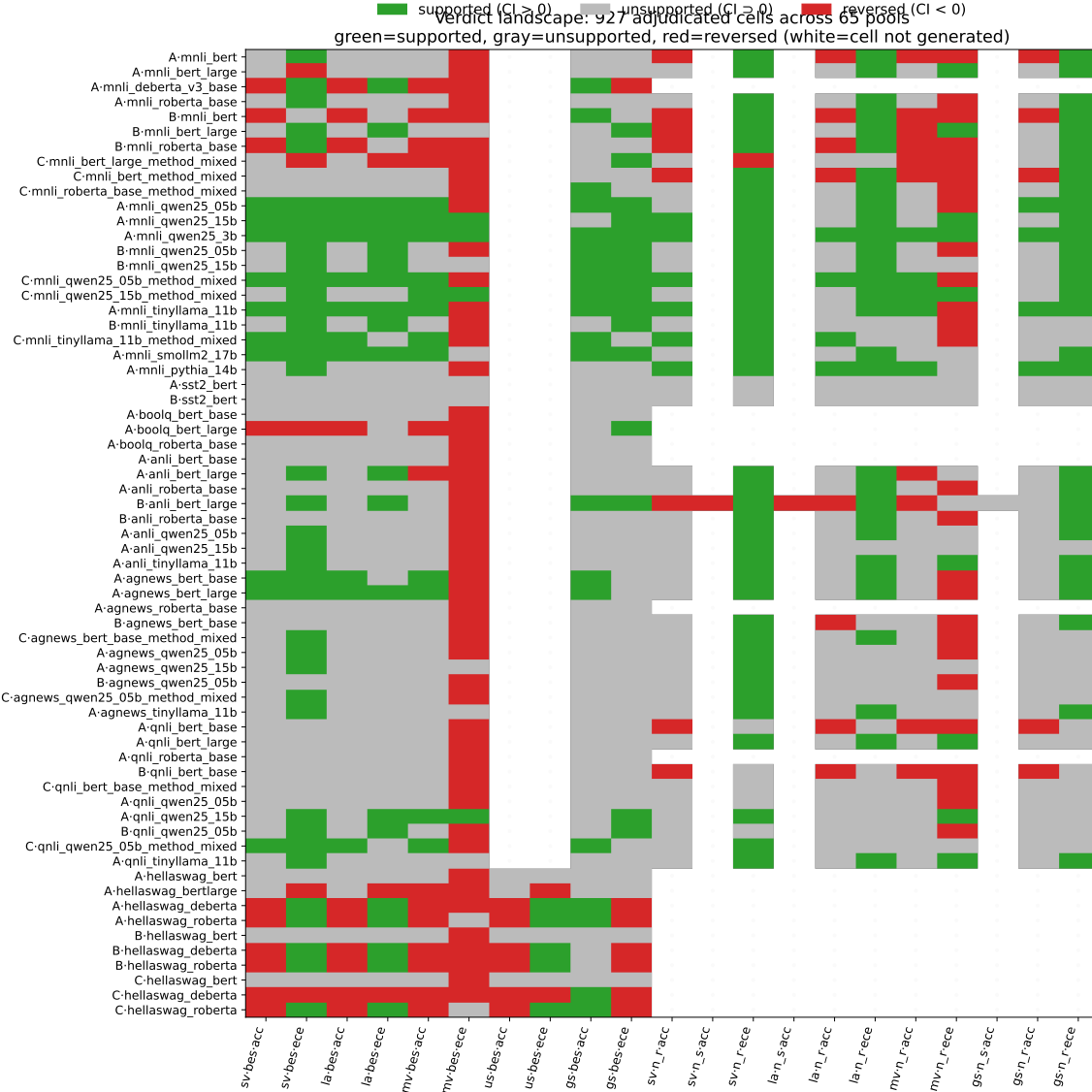


Figure 5: Verdict landscape across the 927 main-pool cells (the remaining 101 baseline-only cells follow the same distribution). Labels reflect the per-cell paired-bootstrap CI *and* corpus-wide BH-FDR ($q=0.05$, cutoff 0.0248): post-FDR 29.5% SUPPORTED / 16.9% REVERSED / 53.6% UNSUPPORTED (pre-FDR 33.0 / 17.8 / 49.2). Encoder accuracy cells are dominantly REVERSED against compute-matched n_{rank} ; decoder cells are more often SUPPORTED. This heterogeneity is itself the evidence that aggregate “ensembles help / don’t help” framings hide structure — and §5 shows the aggregate counts above are themselves protocol-dependent.

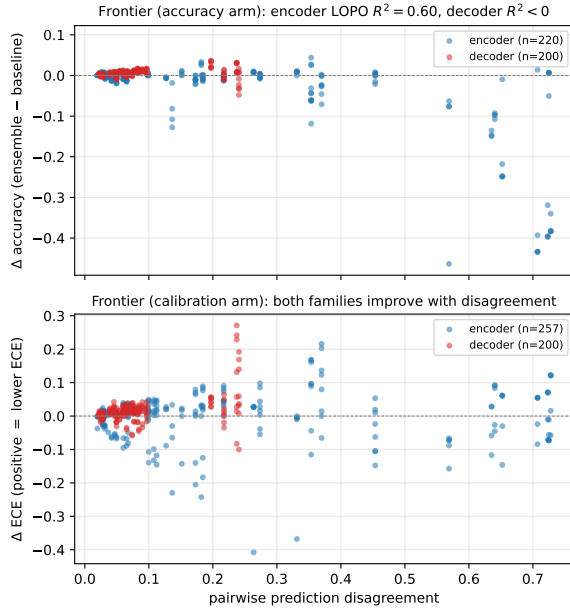


Figure 6: **The diversity-quality frontier and the accuracy arm.** Per-cell scatter of Δ accuracy (top) and Δ ECE (bottom) vs. pairwise prediction disagreement. **(Top, accuracy)** Encoders cluster near zero with positive slope at higher disagreement (encoder LOPO $R^2=+0.60$, Table 2); decoders show no monotone trend (decoder corpus $R^2<0$ at $n=25$ pools, L2). **(Bottom, ECE)** *Family-stratified pattern*: decoders are cleanly separated at the low-ECE tail (consistent with the 21% lower $H(c|\text{conf})$ in App. W); *corpus-wide ECE LOPO is not significant* ($R^2=-0.06$, CI spans 0). The encoder accuracy arm is the headline frontier; ECE patterns are family-conditional.

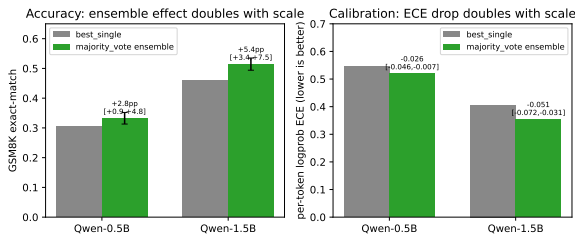


Figure 7: GSM8K decoder scaling on cells with a known selection defect. `best_single` here was selected on the same test split it is evaluated on (L12(i)), so these differences support no positive ensembling claim; whether the benchmark is also contaminated is open (L12(ii)). The embedded panel titles are from an earlier framing and should be read as descriptive, not as a result.